

A Neutrino Carrier Prediction: Robustness and a Framework for Experimental Testing

Cross-paper composition, feature-specific robustness, and held-out evaluation

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Scope

The result applies to the two completed source-qualified vectorlike-joint sectors $\mathcal{B}_8^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}$. It composes the PCR-6 carrier theorem, the PCR-7 charged-current prediction, and the PCR-7.5 robustness and experimental-authority analysis. It introduces no new physical mechanism and makes no claim for the unresolved parent neutrino family, a numerical heavy-particle pole, present-facility sensitivity, or detector-level discrimination without an instantiated SPIC.

This article is a mathematical synthesis of the cited PCR results. The accompanying computational records support provenance and internal consistency; they are not a proof-assistant formalization and do not replace the arguments.

Abstract

This article establishes a child-qualified composition theorem for two completed source-qualified vectorlike-joint neutrino sectors. For every accepted realization, the imported carrier result gives exactly one terminal continuation-bearing hybrid local matter carrier, with three nondegenerate light standing sectors, one rank-three heavy sector, and a continuum-sized quotient of physical realizations. The unperturbed charged-current projection obeys

$$\Delta_{\text{act}} = I_3 - NN^\dagger = V_H V_H^\dagger = \Omega_H = \delta_E I_3, \quad \delta_E > 0,$$

so the inclusive heavy weight equals the active deficit and has rank three. The heavy charged-lepton-flavour-changing amplitude vanishes on the stated common-loop-kernel continuation class.

The contribution of PCR-8 is to show that the carrier, observable, perturbative, and prospective experimental descriptions refer to the same accepted physical apex and descend coherently through their lawful quotients. The robustness statement is feature-specific: one-carrier stability has positive radius on certified trace-transport cells; the closure identity is exact on the block-isometry class; rank three has a positive norm radius; and exact mass degeneracy and flavor universality are independent symmetry-protected properties with zero unrestricted symmetry-breaking radius. Their conjunction is asserted only on the intersection of the corresponding domains.

The ideal carrier/reference separation is structural. Detector-level discrimination is conditional on an experiment-specific SPIC supplying a common record space, response and nuisance models, coverage, resolution, and identifiability bounds. The 28–30 GeV, 0.2 mm pair is therefore retained only as a source-based search benchmark, not as a mass or lifetime prediction. The result remains awaiting qualified held-out evaluation.

A semantic trace classifies finitely presented rivals within the declared same-scope use grammar; it does not claim unrestricted exhaustion of all possible microscopic theories. The unresolved parent family, complete D4 likelihood, universal trace-cell coverage, current-facility sensitivity, numerical complex-pole mass and lifetime, unique microscopic origin, and cross-fermion extensions remain open. Thus the paper provides exact child-qualified prediction closure, feature-specific structural robustness, and a conditional falsification framework without promoting any of those open problems.

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1 Introduction

1.1 Motivation and scope

A multi-stage theoretical programme can fail at its interfaces even when each component result is correct. Adjacent analyses may use different realizations of what appears to be the same scientific object; later conclusions may be silently imported into earlier source definitions; one empirical record may be counted both as an input and as an independent test; or a numerical benchmark may acquire an interpretation not supported by its derivation. The purpose of PCR-8 is to establish that these failures do not occur when the final carrier, observable, robustness, and prospective experimental results are combined.

The positive domain is deliberately narrower than the full neutrino realization space. PCR-5E identifies two completed vectorlike-joint sectors while leaving their parent problem unresolved. PCR-6 classifies terminal carrier identity within each accepted realization. PCR-7 derives the common charged-current observable, and PCR-7.5 determines which carrier and observable properties persist under specified same-scope perturbations [16, 17, 18, 19]. The present paper composes these results without enlarging their domains.

Two distinctions are essential. First,

accepted realization moduli: continuum
 $\neq$ standing spectral sectors: four
 $\neq$ terminal bearers per realization: one.

The one-carrier theorem is a within-realization statement and does not select a unique realization from the continuum-sized quotient. Second,

structural support $\neq$ physical complex pole $\neq$ detector-level identified pole record.

Accordingly, a structural heavy sector is not assigned a numerical mass or lifetime in this article.

1.2 Main contribution

PCR-8 proves that the final PCR-6, PCR-7, and PCR-7.5 objects are compatible over a common accepted realization. It identifies the precise scientific dependencies of the composite argument, establishes lawful quotient descent, and shows that the terminal bearer, observable packet, perturbation data, and conditional record map arise from one physical apex. It also compares the upstream source descriptions before applying endpoint erasure, separates the roles played by empirical inputs, and gives a relative completeness theorem for finitely presented alternatives within the declared same-scope semantic-use class.

The physical conclusion is then obtained by composition rather than by introducing a new neutrino model. For each accepted ready-child realization, there is one terminal carrier and the unperturbed charged-current packet satisfies

$$\Delta_{\text{act}} = \Omega_H = \delta_E I_3, \quad \delta_E > 0, \quad \text{rank } \Omega_H = 3.$$

A prospective discrepancy is localized to a minimal antichain of atomic feature claims, so protocol failure, structural failure, and empirical incompatibility remain distinguishable.

1.3 Feature-specific robustness

Robustness is not represented by one scalar radius. Exact heavy degeneracy is destroyed by arbitrarily small standing traceless perturbations, while exact flavor universality is destroyed by arbitrarily small anisotropic coupling perturbations. Rank three persists on a positive operator-norm ball. The equality $\Delta'_{\text{act}} = \Omega'_H$ is exact on the complete block-isometry class. Terminal carrier identity has a positive radius only on a certified source/deletion/trace-transport cell.

If $U_{\text{car}}^b(X)$ is the certified carrier component, U_{iso} the block-isometry class, and U_{rank} the rank-preserving domain, the strongest joint statement is therefore confined to

$$U_{\text{joint}}^b(X) = U_{\text{car}}^b(X) \cap U_{\text{iso}} \cap U_{\text{rank}}.$$

This intersection condition prevents conclusions established on different perturbation domains from being combined without a common witness.

1.4 Structural prediction and prospective testing

The experimental interpretation has three levels:

structural separation $\Rightarrow$ qualified detector discrimination $\Rightarrow$ empirical confirmation.

The first is a theorem about ideal carrier and reference response classes. The second requires a common record space and an experiment-specific SPIC. The third requires held-out data, independent information ancestry, predeclared nuisance and coverage rules, and a fixed reporting procedure. This paper proves the structural result, states the detector conditions, and defines the held-out evaluation framework. It does not claim current experimental sensitivity or confirmation.

1.5 Organization

Sections 2–5 develop the object system, imported authority, handoff compatibility, and dependency structure. Sections 6–8 study source formation, empirical inputs, and the relative semantic trace. Sections 9–12 give the carrier, observable, robustness, structural-separation, benchmark, and complex-pole results. Sections 13–15 formulate feature-level falsification, open boundaries, the child-qualified composition theorem, and its physical interpretation. Detailed authority, quotient, empirical, robustness, and reproducibility tables are collected in the appendices.

1.6 Limitations

The analysis does not establish that either ready-child sector is realized in nature, select one realization from the continuum-sized quotient, solve the unresolved parent problem, complete THUCL-D3-1.1 or JLR-D4-1, derive a numerical VJ pole mass or proper lifetime, turn the 28–30 GeV benchmark into a predicted mass interval, establish present-facility sensitivity, infer unique microscopic origin from the observable packet, or extend the result to all fermions.

2 Fixed scope, object system, and status calculus

2.1 Ready children and accepted realizations

Define

$$\mathcal{B}_g^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}.$$

For $b \in \mathcal{B}_g^{\text{ready}}$, let $\text{PhysMod}_{E, \text{VJ}}^b$ be the groupoid of accepted source-faithful one-apex physical realizations in that child. An object $X \in \text{PhysMod}_{E, \text{VJ}}^b$ carries the PCR-5E source, target, continuation, failure, and joint-physicality data. PCR-8 never replaces X by a marginal representative. All carrier, observable, perturbation, and protocol objects are attached to the same X or to an explicitly lawful continuation of it.

The child tags are scientific data. The accidental child permits the bounded accidental $B - L$ structure and its nonzero-amplitude D2 stratum; the exact-global child carries the exact-global protector and its target-specific D2/D5 packet. Their common vectorlike source and observable reduct do not identify the decorated child objects.

2.2 Carrier, spectral, and realization levels

For an accepted X , PCR-6 supplies a source-connected parent bearer C_X and a terminal quotient $[C_X]_{\text{term}}$. It also supplies a carrier-bearing spectral core

$$\text{SpecCore}^b(X) = \{S_1, S_2, S_3, S_H\},$$

where S_1, S_2, S_3 are nondegenerate light sectors and S_H is one rank-three heavy sector. The three light sectors remain distinct through oscillation and charged-current witnesses. The heavy rank-one basis presentations collapse under the complete lawful $U(3)_H$ action. Neither the number of displayed eigenvectors nor the number of standing sectors supplies terminal bearer cardinality.

The accepted realization groupoid itself remains non-singleton and continuum-sized. Thus the exact quantifiers are

$$\forall X \in \text{PhysMod}_{E, \text{VJ}}^b : |\text{TermCar}^b(X)| = 1,$$

while

$$|\pi_0 \text{PhysMod}_{E, \text{VJ}}^b| \geq \mathfrak{c}.$$

The first statement is within-realization closure; the second is cross-realization moduli multiplicity.

2.3 Observable and support objects

For each accepted realization, define the unperturbed light and heavy charged-current blocks

$$N = c_E U, \quad V_H = -s_E U, \quad c_E^2 + s_E^2 = 1, \quad \delta_E = s_E^2 > 0,$$

with U unitary. Define

$$\Delta_{\text{act}} = I_3 - N N^\dagger, \quad \Omega_H = V_H V_H^\dagger, \quad C_H = \Delta_{\text{act}} - \Omega_H.$$

The source-qualified heavy domain is $H(I_E)$. PCR-7 used the label $\text{Pole}_H = \text{Deg3}(H_\lambda)$ for the target-visible rank-three heavy support class. PCR-7.5 subsequently separated structural support from unstable-state pole authority. PCR-8 therefore adopts

$$\text{Supp}_H^{\text{str}} := \text{Deg3}(H(I_E))$$

for the structural support/multiplicity object and reserves

$$\text{Pole}_H^{\text{phys}}$$

for the conditional complex-pole construction defined by zeros of the complete inverse propagator.

2.4 Perturbation domains

A same-scope perturbation is a standing-bearing change generated by the frozen PCR-7.5 source grammar, with source updates, target updates, protocol updates, skin, and unresolved cases typed separately. Let

$$\delta K_H = a I_3 + T, \quad T = T^\dagger, \quad \text{tr } T = 0$$

be the heavy mass-squared perturbation and let

$$\eta_V = \|\delta V_H\|_2$$

be the heavy charged-current block perturbation. The following domains are distinct:

$$U_{\text{car}}^b(X) : \text{P75-13-certified carrier trace-transport component,}$$

$$U_{\text{iso}} : \text{complete charged-current block-isometry class,}$$

$$U_{\text{rank}} : \text{rank and heavy-projector stability domain,}$$

$$U_{\text{joint}}^b(X) = U_{\text{car}}^b(X) \cap U_{\text{iso}} \cap U_{\text{rank}}.$$

An accepted unperturbed realization without a certified P75-13 chart still satisfies the exact PCR-6/PCR-7 endpoint, but its positive carrier radius has status $\text{NotReady}_{\text{TraceTransportCell}}$.

2.5 Protocol and record objects

Let $p_\star$ be the finite flavor-complete CCST protocol of PCR-7. Its structural output is a target-qualified record packet. The ideal carrier and reference response classes are $\mathcal{R}_{\text{car}}$ and $\mathcal{R}_{\text{ref}}$. PCR-7.5 introduces a common-record SPIC with record separator $s_{\text{car,ref}}$ and attainable record radius r_{rec} . Same-record selection requires

$$2r_{\text{rec}} < s_{\text{car,ref}},$$

together with the distinct inverse-radius, mass-support, rank/Gram, nuisance, control, sampling, coarsening, and floor guards stated later. The ideal pair-space gap $g_{\text{pair}}^{\text{str}}$ is not a mass gap or a record-space confidence certificate.

2.6 Status classes

PCR-8 uses the following noninterchangeable statuses.

Status	Meaning	Forbidden promotion
ExactUnperturbed	Exact for every accepted ready-child realization at the frozen source	Not a perturbation radius.
ClassExact	Exact throughout a declared perturbation class	Not generic outside that class.
PositiveRadius	Holds on a nonempty open domain with a declared metric	Not universal over uncertified charts.
SymmetryProtected	Exact on a symmetry/equal-singular-value stratum	Zero unrestricted standing-breaking radius.
StructuralGap	Separation in ideal pair space	Not detector confidence.
SPICConditional	Detector-level selection after a complete same-record certificate	Not presently instantiated.
TargetQualifiedBound	Bound inherited from the marked child source	Not independent measurement.
SearchBenchmark	Scientifically motivated scouting point/band	Not a prediction or exclusion.
ConditionalEnvelope	Well-typed specification awaiting named certificates	Not a numerical result.
FutureHeldOut	Predeclared future empirical use	Not current validation.
NotReady	A named burden is missing	Not zero, false, empty, or excluded.
TargetUpdate	The proposal changes the source, target, reference, or protocol identity	Not a same-scope perturbation.

Scope lock

No theorem below may omit the child tag, replace an intersection domain by one of its factors, read structural support as a physical complex pole, call a benchmark a prediction, call an overlapping compatibility use independent validation, or present a conditional SPIC/X7/X8 object as instantiated.

3 Imported results and the unresolved parent

Theorem P8-00. Imported results and scope preservation.

The direct scientific basis consists of PCR-6 v1.3.0, PCR-7 v1.3.0, PCR-7.5 v1.4.4, the original PCR ARC mandate, and the revised PCR-8 theorem specification v0.2.0 [17, 18, 19, 1, 20]. Earlier versions are superseded for direct endpoint use. PCR-5E and the preceding PCR papers enter through the exact roles fixed by the three final handoffs, except where the retrospective analysis examines their formation order. Bibliographic or documentary identity is not used as a scientific premise.

Proof.

The direct-authority table records for each source its programme, title, version, target scope, permitted use, and blocked stronger reading. Earlier direct versions are excluded. Because the PCR-7.5 handoff already carries its transitive scientific dependencies, PCR-8 does not treat them as independent axioms. The four direct sources and the revised theorem specification are unambiguous, and every subsequent use is confined to their stated fields. This establishes source identity and version authority; documentary checks remain evidentiary support rather than a premise of the physical argument.

The resulting status is

$$\text{DirectAuthorityLocked}_8 \oplus \text{NoSupersededEndpointImport}_8.$$

Theorem P8-01. Ready-child scope and the unresolved parent.

The complete positive domain is exactly

$$\mathcal{B}_8^{\text{ready}} = \{b_{\text{acc}}, b_{\text{glob}}\}.$$

No complete PCR-8 carrier/prediction object is formed for the unresolved PCR-5E parent, ROOS components, THUCL-D3-1.1, or the complete D4 target.

Proof.

PCR-6, PCR-7, and PCR-7.5 quantify over the same two source-qualified vectorlike-joint children and each preserves an explicit parent firewall. Their common child reduct therefore defines a functor only on the disjoint union of the two ready child groupoids. A parent object lacks a complete E-H handoff, and the complete D4 target lacks the fresh JLR-D4-1 evaluator. A D3-1.1 object belongs to a distinct target certificate. None can be supplied by interpreting a common child theorem as a covering theorem. Because target identity is part of every object and every restriction map, extending the quantifier would be a target update rather than a consequence. Inspection of the positive theorem statements confirms that each either contains $b \in \mathcal{B}_8^{\text{ready}}$ or is an explicitly child-independent authority statement.

Hence

$$\text{ParentNotReady}_8, \quad \text{D3-1.1Separate}_8, \quad \text{D4CompleteNotReady}_{\text{JLR-D4-1}}.$$

3.1 Why common child structure does not close the parent

A common reduct is an agreement theorem on the overlap of two completed target objects. It is not a descent theorem from their union to an unresolved ambient target. The parent may contain source kinds, target interactions, response histories, or overlap components absent from both children. Any universalization must therefore construct its own parent target, source category, complete handoff, and restriction proof. PCR-8 records that future route but does not simulate it by widening notation.

4 Typed three-handoff compatibility and cross-paper object identity

4.1 The common accepted-realization index and identity groupoid

For each $b \in \mathcal{B}_8^{\text{ready}}$, let $\mathcal{A}_b$ be the groupoid of accepted source-faithful physical apices $X \in \text{PhysMod}_{E, \text{VJ}}^b$ and their lawful physical isomorphisms. Let $\mathcal{I}_b$ be the groupoid whose objects are

$$\iota = (b, X, S_X, s_E, c_E, H(I_E), \text{Cont}_X, \partial_b).$$

Here S_X is the activity-reduced source/physical reduct, Cont_X is the licensed continuation action retained by all three handoffs, and ∂_b is the complete child/parent and target-update boundary. An arrow of $\mathcal{I}_b$ is a lawful source/physical isomorphism preserving all eight fields. The functor

$$j_b : \mathcal{A}_b \longrightarrow \mathcal{I}_b$$

retains this common identity record.

Write $\mathcal{H}_6^b$, $\mathcal{H}_7^b$, and $\mathcal{H}_{7.5}^b$ for the final PCR-6, PCR-7, and PCR-7.5 handoff groupoids. They have forgetful functors

$$\pi_6 : \mathcal{H}_6^b \rightarrow \mathcal{I}_b, \quad \pi_7 : \mathcal{H}_7^b \rightarrow \mathcal{I}_b, \quad \pi_{7.5} : \mathcal{H}_{7.5}^b \rightarrow \mathcal{I}_b.$$

4.2 Exact upstream binding of the section functors

The three sections are not inferred from similar notation. Their object and arrow actions are bound to the exact final handoff theorems listed below.

Functor	Object action	Arrow action	Exact source binding	Naturality witness
s_6	attach the P6-26 terminal/nature packet to X	P6-R2b moduli transport of sector, bearer, and continuation data	PCR-6 v1.3.0 P6-R2b, P6-R5, and P6-26	$\epsilon_6 : \pi_6 s_6 \Rightarrow j_b$
s_7	attach the P7-28 packet, protocol, reference, and falsifier to the same X	P7-08 lawful record naturality on accepted-child arrows	PCR-7 v1.3.0 P7-08, P7-R4, and P7-28	$\epsilon_7 : \pi_7 s_7 \Rightarrow j_b$
s_{75}	attach the P75-H1–H10 robustness and authority packet	P75 perturbation/status transport preserving source and boundary fields	PCR-7.5 v1.4.4 P75-22, P75-23, and P75-26/Theorem 19.1	$\epsilon_{75} : \pi_{75} s_{75} \Rightarrow j_b$

These exact theorem and version bindings show that the sections are scientific exports of the final upstream papers rather than locally chosen identifications.

Theorem P8-02. Typed three-handoff homotopy pullback.

For every $b \in \mathcal{B}_8^{\text{ready}}$, the iterated iso-comma groupoid

$$\mathcal{H}_8^b := \mathcal{H}_6^b \times_{I_b}^h \mathcal{H}_7^b \times_{I_b}^h \mathcal{H}_{7.5}^b$$

is defined and nonempty over every accepted X . Its objects are

$$(h_6, h_7, h_{75}; \beta_{67}, \beta_{7,75}),$$

where $\beta_{67} : \pi_6 h_6 \xrightarrow{\sim} \pi_7 h_7$ and $\beta_{7,75} : \pi_7 h_7 \xrightarrow{\sim} \pi_{75} h_{75}$. Its arrows are triples of handoff arrows for which the two β -squares commute. Conditional and NotReady decorations retain their status. The groupoid has the standard homotopy-pullback universal property.

Proof.

The preceding source-binding table supplies actual functors on objects and arrows and the natural isomorphisms

$$\epsilon_6 : \pi_6 s_6 \Rightarrow j_b, \quad \epsilon_7 : \pi_7 s_7 \Rightarrow j_b, \quad \epsilon_{75} : \pi_{75} s_{75} \Rightarrow j_b.$$

For $X \in \mathcal{A}_b$, set

$$\alpha_{67}(X) = \epsilon_7(X)^{-1} \epsilon_6(X), \quad \alpha_{7,75}(X) = \epsilon_{75}(X)^{-1} \epsilon_7(X).$$

Then

$$(s_6 X, s_7 X, s_{75} X; \alpha_{67}(X), \alpha_{7,75}(X))$$

is a pullback object, proving nonemptiness without selecting a new representative.

Let $\mathcal{K}$ carry functors f_6, f_7, f_{75} into the three handoff groupoids and coherent isomorphisms $\pi_6 f_6 \cong \pi_7 f_7 \cong \pi_{75} f_{75}$. The iso-comma construction gives a functor $\mathcal{K} \rightarrow \mathcal{H}_8^b$ objectwise and arrowwise; any second factorization with the same cone is connected by the canonical componentwise natural isomorphism. This is the universal property. The pullback retains the PCR-6 terminal object, PCR-7 packet/protocol/reference, and PCR-7.5 robustness and authority decorations while leaving SPIC instantiation and X7/X8 numerical poles conditional.

HandoffPullbackClosed₈.

Theorem P8-04. Cross-paper scientific-object identity.

Every PCR-8 target, source, realization, carrier, spectral sector, observable, perturbation, protocol, reference, record, gap, and status field has a stable scientific identity across $\mathcal{H}_8^b$. In particular, the PCR-7 legacy label Pole_H is normalized to

$\text{Supp}_H^{\text{str}}$, while $\text{Pole}_H^{\text{phys}}$ remains a distinct X7/X8 object. The ideal pair-space gap $g_{\text{pair}}^{\text{str}}$, structural mass-support gap $g_{\text{mass}}^{\text{str}}$, observed reconstructed interval gap $\widehat{g}_{\text{mass}}^{\text{str}}$, same-record separator $s_{\text{car,ref}}$, and confidence radius r_{rec} are pairwise typed.

Proof.

The object crosswalk records formation source, codomain, lawful relation, and blocked conflation. Literal equality is used only for common identity fields. Projection is used for $X \mapsto (N, V_H) \mapsto (\Delta_{\text{act}}, \Omega_H)$; restriction is used from a decorated child to its common charged-current reduct. The legacy support label receives an authority-narrowing rename, while $\text{Pole}_H^{\text{phys}}$ requires the separate complex-pole construction. The five gap/radius objects inhabit ideal pair space, true mass-support space, reconstructed mass-interval space, and common detector-record space. No identity follows from symbol or numerical coincidence.

The normalized exact packet is

$$O_{\text{core}}(X) = ([C_X]_{\text{term}}, \Delta_{\text{act}}, \Omega_H, \text{Supp}_H^{\text{str}}),$$

not a packet containing $\text{Pole}_H^{\text{phys}}$ or a detector confidence radius.

4.3 Summary of handoff relations

Field	Primary source	PCR-8 relation	Status
Common identity index	accepted physical apices	$\mathcal{A}_b \xrightarrow{j_b} \mathcal{I}_b$	exact
Handoff comparison	final handoffs	$\epsilon_i : \pi_i s_i \Rightarrow j_b$	exact/source-bound
Capstone handoff	P8-02	homotopy pullback with universal property	exact
Terminal bearer	PCR-6	retained parent bearer C_X	exact
Light/heavy blocks	PCR-7	source projection of the same X	exact
CCST packet	PCR-7	natural record of X	exact unperturbed
Perturbation trace	PCR-7.5	standing decorations of the frozen source	exact classification
Carrier radius	PCR-7.5	positive on declared trace cells	cell-certified
Pair-space gap	PCR-7.5/P8-16	$U_{\text{iso}} \cap U_{\text{rank}}$	exact structural
Record selection	PCR-7.5	common-record SPIC	conditional
Search benchmark	PCR-7.5 X1–X4	experiment-map layer	benchmark
Weak normalization	PCR-7.5 X5	marked-child bound and screens	mixed exact/NotReady
Physical complex pole	PCR-7.5 X7/X8	separate continuation object	conditional/NotReady numerically
Parent	PCR-5E/PCR-6	explicit absent handoff	NotReady

5 Scientific dependencies, quotient descent, and one-apex composition

5.1 Scientific dependencies and proof-use structure

PCR-8 distinguishes three complementary dependency structures.

1. $G_{8,\text{sci}}^{\text{reg}}$ is the scientific object graph determined by the stated formation relations.
2. $G_{8,\text{sci}}^{\text{actual}}$ records explicit theorem and object references in the P8 proofs; $G_{8,\text{sci}}^{\text{decl}}$ records their theorem-local direct uses; and the

independently reviewed $G_{8,sci}^{inv}$ records the same dependencies from the source side.

3. G_8^{math} records ambient mathematical results—iso-comma universal properties, quotient factorization, finite-poset antichains, matrix perturbation, and finite-sample inequalities—without treating them as scientific PCR stages.

For a P8 theorem w , let $\Pi_{reg}(w)$ denote its scientific inputs, $\Pi_{decl}(w)$ its stated direct uses, and $\Pi_{actual}(w)$ the explicit references appearing in its proof. The ambient mathematical authorities form the separate list $\Pi_{math}(w)$. These descriptions are formed independently and compared only afterward.

Theorem P8-03. Proof-use reconciliation and dependency structure.

The registered scientific graph $G_{8,sci}^{reg}$ is acyclic. For every P8 theorem w ,

$$\Pi_{actual}(w) \subseteq \Pi_{decl}(w) = \Pi_{inv}(w) = \Pi_{reg}(w),$$

and every $m \in \Pi_{math}(w)$ is bound to a separately registered ambient mathematical authority. Hence the scientific dependency graph is not inferred from the internal consistency of a single description: every explicit proof reference must occur among the declared and scientific uses, while the complete direct-use set is reconciled with the independent inventory and graph.

Proof.

Each scientific edge records its source theorem, exported object, target, scope, use in the receiving theorem, and blocked stronger reading. Independently, every P8 proof identifies its direct scientific and mathematical sources. The explicit references, including PCR-2 target/interface uses, are compared with the theorem-local declarations, source-side inventory, and incoming graph edges. An undeclared reference, an unregistered use, or a load-bearing edge without a corresponding theorem use would violate the reconciliation.

The upstream scientific results are not re-proved here. Their authority is fixed by the exact source theorems and the source signatures used in P8-07. Ambient mathematics is considered separately so that a categorical universal property, perturbation inequality, or finite-poset result is not mistaken for a scientific PCR stage. Every scientific edge strictly increases the independently assigned formation rank; acyclicity follows by topological ordering only after the edge set has been reconciled.

Thus two distinct facts follow:

$$G_{8,sci}^{actual} \subseteq G_{8,sci}^{decl} = G_{8,sci}^{inv} = G_{8,sci}^{reg} \quad \text{on local P8 proof uses,} \quad G_{8,sci}^{reg} \text{ is acyclic,}$$

while G_8^{math} remains a typed authority layer rather than hidden scientific ancestry.

IndependentProofUseReconciliationClosed₈ $\oplus$ ScientificDependencyDAGClosed₈.

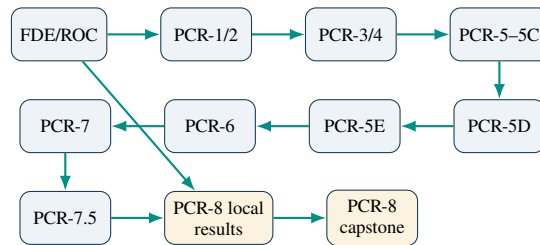


Figure 1: Projection of the complete theorem and scientific-object dependency graph.

5.2 Quotient objects and explicit descended maps

Let q_{src} , q_{phys} , q_{SCIC} , q_{acc} , q_{term} , q_{obs} , q_{frozen} , and q_{pert} denote the lawful quotient maps at the corresponding source, physical, standing, accepted, terminal, observable, and perturbation levels. A detector record quotient q_{rec}^P is formed only after an instantiated protocol-specific SPIC.

Theorem P8-05. Lawful quotient descent and redescription diagram.

The exact structural maps are congruent with their lawful equivalence relations and therefore admit unique descended maps satisfying

$$\begin{aligned} q_{\text{phys}} \circ F_{\text{phys}} &= \overline{F}_{\text{phys}} \circ q_{\text{src}}, & q_{\text{SCIC}} \circ F_{\text{scic}} &= \overline{F}_{\text{scic}} \circ q_{\text{phys}}, \\ q_{\text{term}} \circ F_{\text{term}} &= \overline{F}_{\text{term}} \circ q_{\text{acc}}, & q_{\text{obs}} \circ F_{\text{obs}} &= \overline{F}_{\text{obs}} \circ q_{\text{acc}}, \\ q_{\text{pert}} \circ \text{Pr}_{\text{pert}} &= \overline{\text{Pr}}_{\text{pert}} \circ q_{\text{frozen}}. \end{aligned}$$

The terminal and observable functors are parallel consequences of the same accepted physical apex; neither is inferred from the other. Under an instantiated SPIC, and only then,

$$q_{\text{rec}}^p \circ F_{\text{rec}}^p = \overline{F}_{\text{rec}}^p \circ q_{\text{obs}}^p.$$

Proof.

PCR-3 source equivalence preserves the activity-reduced carrier trace, so source-faithful physical realization is congruent. PCR-5B/SCIC equality is complete standing-continuation-signature equality, so lawful physical isomorphism descends to SCIC. PCR-6 transports the terminal interface, support graph, eligibility, compatibility, and unique assignment orbit under lawful accepted-apex isomorphism. PCR-7 naturality gives observable congruence. PCR-7.5 gives perturbation-trace congruence on the frozen source/observable base. In each case define the descended map by $\overline{F}([x]) = [F(x)]$; congruence gives representative independence, the displayed square commutes, and surjectivity of the quotient map gives uniqueness. SPIC supplies the additional common-record congruence needed by the last square. Without SPIC the detector leg is not a total descended map.

QuotientDescentClosed₈.

The exact nonidentifications are

$$\begin{aligned} &\text{source presentation} \neq \text{carrier identity} \neq \text{spectral sector}, \\ \text{Supp}_H^{\text{str}} &\neq \text{Pole}_H^{\text{phys}}, & g_{\text{pair}}^{\text{str}} &\neq g_{\text{mass}}^{\text{str}} \neq \widehat{g}_{\text{mass}}^{\text{str}} \neq s_{\text{car,ref}} \neq r_{\text{rec}}. \end{aligned}$$

Theorem P8-06. One-apex carrier-to-observable composability.

For every $b \in \mathcal{B}_8^{\text{ready}}$ and accepted X , the terminal bearer, internal spectral core, charged-current packet, perturbation trace, and target-qualified protocol are evaluated on the same source-faithful physical apex or an explicit lawful continuation of it. The retained coherence web consists of source–apex, apex–terminal, apex–observable, terminal/observable common-identity, observable–perturbation, perturbation–record, formation-ancestry, child-boundary, and common-cone cells. No marginal splicing is used.

Proof.

PCR-5E supplies one-apex joint physicality. PCR-6 constructs C_X from the source-connected neutral-fermion core and retains every deletion-essential relation. PCR-7 derives N, V_H and the packet from the same source map. PCR-7.5 decorates the frozen source and protocol rather than replacing X by marginal fits. P8-02 gives the coherent handoff cone, and P8-05 gives representative-independent terminal, observable, perturbation, and conditional record maps. The eleven specified relation cells therefore commute on one composite object, including the physical and terminal actions under perturbation and their compatibility with observable transport and the child boundary. Child-specific D2/D5 decorations remain present but do not enter the common charged-current reduct without an explicit backreaction theorem.

OneApexCompositionClosed₈.

6 Source formation and endpoint-erasure analysis

6.1 Source descriptions and formation order

For each load-bearing upstream stage k , consider two independently prepared descriptions.

1. $\text{Sig}_{\text{src}}^{\text{NF}}(k)$ is a source-normal-form signature: a normalized formation record containing the source identity, theorem or definition pointer, target, comparison class, primitive input fields, direct dependencies, their coverage relation, and the resulting object. It is not a literal reproduction of the source theorem's syntax.
2. $\text{Sig}_{\text{audit}}(U_k)$ is the corresponding input description used in the endpoint-erasure argument, prepared independently from the source-normal-form account.

The comparison is made at the level of fields, targets, dependencies, and source pointers. Only after

$$\text{Sig}_{\text{src}}^{\text{NF}}(k) = \text{Sig}_{\text{audit}}^{\text{NF}}(U_k)$$

is established do we form

$$D_k = U_k \times L_k, \quad E_k = \text{pr}_{U_k} : D_k \rightarrow U_k,$$

and the source-bound maps

$$\text{Form}_k^{\text{src}} : U_k \rightarrow O_k, \quad F_k := \text{Form}_k^{\text{src}} \circ \text{pr}_{U_k}, \quad \tilde{F}_k := \text{Form}_k^{\text{src}}.$$

The commuting equality $F_k = \tilde{F}_k \circ E_k$ is therefore the final step, not the evidence for the source signature.

Theorem P8-07. Source-reconciled endpoint-erasure analysis.

For every load-bearing formation stage from PCR-1 through PCR-7.5, the source signature and audit signature agree at field resolution; all later fields are absent from the source signature; and the source-bound formation map factors through endpoint erasure. Therefore

$$\text{NoLoadBearingEndpointDependence}_8.$$

The separate historical status remains

$$\text{HistoricalProvenanceStatus}_8 = \text{EvidenceBounded_NotCompletePsychologicalCertification}.$$

Proof.

The source-normal-form comparison covers fifteen separately typed formation stages. Each row identifies the direct source theorems or definitions and the primitive formation fields, and relates every dependency to at least one normalized field. Losslessness here means that this relation is total on the enumerated direct dependencies, surjective onto the normalized fields, and preserves the target, comparison class, resulting object, and equality type. PCR-1 through PCR-4 use the exact target, interface, source, survivor, and common-cone definitions named in the table below. PCR-5B explicitly enumerates B00–B15, rather than treating seven aggregate labels as the literal B15 signature; PCR-5C uses C01–C20 and Theorem 17.2. D1–D5 are no longer compressed into one audit row: each carries its own target certificate, source extension, one-apex realization, retained restriction, failure boundary, and final handoff. The ready-child D3 row is explicitly the successor target THUCL-D3-1.3, while D4 retains the public-summary versus complete-likelihood boundary. PCR-5E uses E15/E-H1–E-H10; PCR-6 uses P6-R2b/P6-R5/P6-26; PCR-7 uses P7-08/P7-R4/P7-28; and PCR-7.5 uses P75-23/P75-26. The independent endpoint-erasure description lists the same categories without being derived from the source-side normal form. Their comparison rejects any uncovered dependency or normalized field, any missing or additional field, and any mismatch in target, comparison class, version, or resulting object.

After reconciliation, the source theorem supplies $\text{Form}_k^{\text{src}}$. The extension to D_k ignores L_k by projection, giving the displayed triangle literally or through the exact quotient/groupoid isomorphism named by the source. Each negative control inserts one later selector or replaces one source target; the result fails the source signature or output-identity comparison before the triangle is accepted. The fifteen rows cover every upstream formation family used in the reconciled proof dependency structure, including each D-series target separately.

This establishes semantic nondependence. It does not establish that later ideas were absent from every historical thought process; that stronger proposition remains evidence-bounded.

6.2 Comparison of source descriptions

Stage	Exact source locator	Independently reconciled object	Equality type	Status
PCR-1	Def. 2.14; Thms. 2.15, 13.1	target fingerprint and H01–H10	literal/lawful quotient	pass
PCR-2	Defs. 3.2–3.3; Thms. 13.12–13.13, 17.1	target trace/use/interface	canonical interface iso	pass
PCR-3	Defs. 6.4–6.5; Thms. 17.3–17.5, 17.9	source/activity/candidate universe	carrier-semantic iso	pass
PCR-4	Thm. 3.2; Def. 7.4; Thms. 7.5, 21.1–21.2	one-apex survivor relation	survivor-quotient iso	pass
PCR-5B	B00/Thm. 2.1; B15/Thm. 19.1	exact base atlas/B15 interface	groupoid equivalence	pass
PCR-5C	C01–C20; Thm. 17.2	restriction/lift/joint calculus	pseudofunctor package	pass
PCR-5D1	D1-01; source compiler; D1-04; D1-08	KATRIN/source/restriction handoff	target-tagged naturality	pass
PCR-5D2	D2-01/02; Thms. 6.4, 6.6; D2-04/12	exact-amplitude D2 handoff	target-tagged naturality	pass
PCR-5D3-1.3	STX-00/01/04/08/23/24	successor active-closure handoff	qualified restriction image	pass
PCR-5D4	Defs. 3.2–3.3; Thms. 3.4–3.5; D4-12	public-summary D4 handoff/boundary	summary-target equality	pass
PCR-5D5	D5-01/02/03/04/12	symmetry/charge marginal interface	target-indexed equality	pass
PCR-5E	Def. 17.1; Thm. 17.2/E15	E-H1–E-H10 children/parent	child-handoff equality	pass
PCR-6	P6-R2b; P6-R5; P6-26	bearer/nature handoff	moduli-natural iso	pass
PCR-7	P7-08; P7-R4; P7-28	packet/protocol/reference handoff	record naturality	pass
PCR-7.5	P75-23; P75-26/Thm. 19.1	robustness/authority handoff	status-vector equality	pass

6.3 Exclusion of endpoint leakage

A faithful defeat identifies a stage k and later field $z \in L_k$ for which the exact source signature contains z or the source formation cannot be reconstructed from the reconciled U_k while preserving the same target and comparison class. Editorial foresight is not sufficient; conversely, a failed source reconciliation may not be dismissed as presentation.

7 Empirical inputs, parameter authority, and evidentiary limits

7.1 Why empirical authority is use-level

A data set can supply more than one scientific use. One subset may form a target, another may calibrate a response, and a third may serve as a consistency check. Assigning one role to an entire file hides rather than resolves information overlap. PCR-8 therefore defines an empirical use as

$$\text{EmpUse} = (\text{dataset/version, subrecord, theorem use, coordinate, role}).$$

Held-out eligibility is a property of the information ancestry of a particular use, not of a filename.

Theorem P8-08. Empirical-use classification, overlap, and held-out separation.

Every empirical use in the capstone has an exact role and overlap ancestry. No anchor, calibration, derivation, benchmark, or overlapping compatibility use is counted as independent validation of the same coordinate. A future record is held-out eligible only if its information ancestry is disjoint from the upstream formation uses, its protocol matches $p_\star$ or a pre-certified refinement, its nuisance/covariance model is frozen before unblinding, and its report uses the preregistered packet and falsifier without retuning.

Proof.

The empirical-use classification separates oscillation, KATRIN, ^{76}Ge , D3, D4-summary, D5, active-tester, ATLAS-grid, HFLAV, W -mass-screen, future CCST, future LNV, complete-cosmology, and future X7/X8 uses. The overlap graph contains a directed ancestry edge from the D3 LFU block to the HFLAV compatibility use. Therefore HFLAV is not statistically independent validation of that child coordinate. The ATLAS grid and its ordinate replay share one search-response source and cannot validate each other. The future flavor-complete packet is eligible only for the predeclared relation not already used in child formation and only after SPIC qualification. These rules exactly match the PCR-7 held-out theorem and PCR-7.5 empirical firewall. The classification has no role collision capable of manufacturing independence. Hence the data firewall closes.

DataFirewallClosed₈.

7.2 Parameter authority lattice

The capstone displays numbers only after assigning their authority. The lattice is ordered by what the number can lawfully do, not by precision:

StructuralExact, TargetQualifiedBound, SourceCoordinate, SearchBenchmark,
CompatibilityCheck, ScreeningOnly, ConditionalEnvelope, ProjectionOnly, NotReady.

A later position in this display is not weaker in every sense; it is differently authorized.

Theorem P8-09. Parameter authority and numerical non-substitution.

The following statuses are exact:

$$0 < \epsilon_{\text{row}} = \delta_E = s_E^2 \leq \frac{1}{256} \quad \text{is a marked-child target-qualified source bound,}$$

$H(I_E)$ is a symbolic source-qualified heavy domain, the D3 pair ([28, 30] GeV, $c\tau_A^{\text{resp}} = 0.2$ mm) is a search benchmark, the HFLAV statistic is an overlapping compatibility check, $\epsilon_{95}^{\text{scr}}$ is screening-only, and $\epsilon_{95}^{\text{weak}}$ is NotReady. Structural support and physical complex-pole mass/lifetime are separate.

Proof.

Each displayed parameter is traced to the map that licenses it. The row coefficient is inherited from the marked child source and is neither a likelihood estimate nor a target-independent selector. The source domain $H(I_E)$ is kept symbolic by PCR-7.5; no numerical endpoints are imported. The D3 benchmark is licensed by the source-certified continuum, target margins, direct 30 GeV anchor, and discrete 0.2 mm response-grid point, not by a mass-ratio theorem. The HFLAV calculation has $p = 0.1349$ and zero common- ϵ information. The separate W -mass leading-response calculation produces $\epsilon_{95}^{\text{scr}} \approx 1.62 \times 10^{-3}$ but lacks the complete physical coverage required for $\epsilon_{95}^{\text{weak}}$. The X7/X8 schema distinguishes $\text{Supp}_H^{\text{str}}$ from $\text{Pole}_H^{\text{phys}}$. The parameter analysis therefore excludes all five laundering routes: source coordinate to measurement, benchmark to prediction, symbolic domain to numerical interval, response coordinate to proper lifetime, and screen to covered physical endpoint.

NoParameterLaundering₈.

Theorem P8-10. Evidentiary non-substitution and proof authority.

Documentary and computational checks establish identity, scope consistency, and reproducibility. They do not prove target completeness, source completeness, one-apex physicality, terminal identity, observable necessity, protocol realizability, or empirical truth.

Proof.

The claim-authority correspondence assigns each scientific proposition to a theorem in this manuscript or an upstream paper, while reproducibility statements are supported by documentary and computational evidence. These forms of evidence meet only at consistency: a reproducibility record may confirm that a theorem statement, target, or status has been transmitted faithfully, but it cannot establish the physical semantics of that statement. Conversely, a semantic proof does not by itself establish the integrity or visual fidelity of a distributed copy. The scientific and reproducibility arguments are therefore complementary and noninterchangeable.

CertificateNonSubstitution_g.

7.3 Summary of empirical roles

Use	Role	Capstone use	Held-out status
Oscillation evidence	anchor	base target	no
KATRIN/ ⁷⁶ Ge/D3/D4	anchor/calibration/derivation	child formation and controls	no
ATLAS response grid	benchmark/screen	scouting architecture	no
HFLAV	overlapping compatibility check	stop-rule validation only	no
<i>W</i> -mass screen	screening-only	numerical leverage, no promotion	no
Future flavor-complete packet	future held-out	primary CCST evaluation	conditional yes
Future independent LNV	future held-out	one-sided child discriminator	conditional yes
Complete cosmology	FutureNotReady	target update	no current use
X7/X8 response	FutureNotReady	mass/lifetime authority	no current use

8 Semantic capstone uses and relative rival classification

8.1 Definition of the same-scope use class

Fix a ready child b . A raw capstone presentation Y contains the accepted target/source, physical apex, terminal interface, observable projection, perturbation/update data, and conditional protocol/record data, together with the maps by which those objects refer to one physical realization. Define $\text{Use}_{\text{cap}}^b$ before defining any capstone trace as the saturation of the following independently authorized use families:

1. PCR-2 target/interface definedness and lawful-redescription uses;
2. PCR-3 source-identity, activity, and finite source-query uses;
3. PCR-4/PCR-5E one-apex, residual, overlap, and common-cone uses;
4. SCIC/PCR-6 terminal standing, continuation, eligibility, compatibility, and ROC uses;
5. PCR-7 semantic observable, protocol, reference, and falsifier uses;
6. PCR-7.5 perturbation, feature-domain, update, SPIC-status, and authority uses;
7. lawful finite composition, restriction, continuation, failure localization, and postprocessing of the preceding uses.

A semantic capstone use is fail-closed, same-target, invariant under lawful source/physical/record isomorphism, and contains no extra ranking or selector. This class is formed from the frozen interfaces and operations, not from the desired endpoint and not from equality of a proposed trace. Define semantic capstone equivalence by

$$Y_1 \equiv_{\text{cap}} Y_2 \iff \nu(Y_1) = \nu(Y_2) \text{ with equal definedness for every } \nu \in \text{Use}_{\text{cap}}^b.$$

8.2 Derived trace and eleven coherence relations

The primitive restrictions of the independent use class define

$$\text{Tr}_8(Y) = (A_Y, P_Y, T_Y, O_Y, U_Y, R_Y; C_Y),$$

with the six object layers used throughout the manuscript. The coherence package is finite and has no catch-all relation:

Cell	Exact commuting requirement
C_{AP}	$\text{SrcRed}(P_Y) = A_Y^{\text{src}}$ and the source-activity restriction commutes with lawful arrows.
C_{PT}	the terminal restriction and ROC evaluation factor through the same apex: $T_Y = \text{Term}(P_Y)$.
C_{PO}	the observable object is the source/physical projection of that apex: $O_Y = \text{Obs}(P_Y)$.
C_{TOI}	$\pi_T(T_Y) \cong j_b(A_Y) \cong \pi_O(O_Y)$ in $\mathcal{I}_b$; no terminal-to-observable generator is inferred.
C_{PU}	for every licensed same-scope perturbation u , $\text{Phys}(uY) \cong u_P \text{Phys}(Y)$, naturally in lawful apex arrows and with the ready-child boundary fixed.
C_{TU}	for every licensed same-scope perturbation u , $\text{Term}(uY) \cong u_T \text{Term}(Y)$, including terminal support, ROC, eligibility, compatibility, and assignment-orbit transport.
C_{OU}	for every licensed same-scope perturbation u , $\text{Obs}(uY) = u_O \text{Obs}(Y)$.
C_{UR}	on an instantiated protocol leg, $\text{Rec}(uY) = u_R \text{Rec}(Y)$ with status/coverage typing preserved.
C_{AR}	each formation-ancestry square is the source-reconciled triangle of P8-07, not a record-derived selector.
C_∂	restriction to b and routing to the unresolved parent/update complement commute with A, P, T, O, U, R .
C_Δ	the identity, physical/terminal/observable perturbation, record, and boundary diagrams agree on overlaps: $C_{TU} \circ C_{PT} = C_{PT}^u \circ C_{PU}$, $C_{OU} \circ C_{PO} = C_{PO}^u \circ C_{PU}$, $\pi_T C_{TU}$ and $\pi_O C_{OU}$ agree through C_{TOI} , and $C_{UR} \circ C_{OU} \circ C_{PO}$ is the record projection of the same perturbed apex. These squares commute with C_∂ and their induced higher cocycle is the identity.

Thus C_Δ is a displayed finite cocycle condition; it is not a container for unlisted relations.

Theorem P8-11A. Completeness relative to the declared capstone-use class.

For every $\nu \in \text{Use}_{\text{cap}}^b$ there is a lawful partial functional $\widehat{\nu}$ such that

$$\nu(Y) = \widehat{\nu}(\text{Tr}_8(Y)).$$

Conversely, a difference in any trace layer or any of the eleven displayed coherence cells is witnessed by an admitted primitive capstone use. Consequently

$$Y_1 \equiv_{\text{cap}} Y_2 \iff \text{Tr}_8(Y_1) \cong \text{Tr}_8(Y_2).$$

Within this declared use class, every finitely presented same-scope rival differs first in a specified trace factor or cell or is capstone-equivalent; a trace-preserving contrary endpoint remains a genuine counterexample to this theorem or to its first false premise.

Proof.

The use class is formed independently of the trace. Prove descent by structural induction over its seven generator families. PCR-2 target uses factor through A and the target portion of P ; PCR-3 source uses factor through A and C_{AP} ; PCR-4/PCR-5E simultaneous uses factor through P , C_{AP} , and the explicit overlap/cocycle cells; SCIC/PCR-6 terminal uses factor through T , C_{PT} , and C_{TOI} ; PCR-7 observable uses factor through O , C_{PO} , and the reference relation. PCR-7.5 perturbation and authority uses act simultaneously through C_{PU} , C_{TU} , and C_{OU} on the physical, terminal, and observable layers, with C_{Δ} proving that terminal-after-perturbation and observable-after-perturbation are evaluations of the same perturbed apex. The action is boundary-preserving through C_{∂} . Conditional record uses factor through R , C_{UR} , and C_{AR} . Composition and continuation preserve factorization because the action squares and higher cocycle commute.

For soundness, each primitive trace field is itself the output or definedness status of one independently admitted use. Each coherence cell is tested by the corresponding naturality, common-identity, ancestry, boundary, or cocycle use. Hence unequal traces are semantically distinguishable. For saturation, PCR-3 finite source completeness, PCR-4 relation/common-cone completeness, PCR-6 terminal-use descent, PCR-7 semantic observable-use completeness, and PCR-7.5 perturbation/update totality cover the seven generators; Fixed-Domain Exhaustion blocks an additional same-scope authorizer after those target, quotient, continuation, compatibility, failure, and boundary loci are fixed. Capstone equivalence was defined by all semantic uses, not by trace isomorphism. The biconditional is therefore a derived completeness theorem rather than a restatement of a definition.

$$\text{SemanticCapstoneUseComplete}_8 \oplus \text{CapstoneTraceComplete}_8 \oplus \text{RivalNormalFormExhausted}_8.$$

Theorem P8-11B. Rival classification within the declared same-scope class.

Every rival receives its complete minimal difference set and one of the statuses

SamePresentation, DifferentSourceOrTarget, OneApexFailure,
TerminalIncomplete, CrossInterfaceCoherenceFailure, SameCarrierPerturbation,
ObservableSeparated, ObservationalMimic, ProtocolNotQualified,
ParentNotReady, GenuineCounterexample.

A frozen reporting priority may select a primary label only after the co-minimal difference set is retained.

Proof.

Apply P8-11A to the semantic use class. The no-vectorlike reference is structurally separated and only conditionally record-separated under SPIC. A proposal lacking the ready-child source/target fingerprint is a different target; one lacking the terminal signature is terminal-incomplete. Agreement of all six object layers with failure of one of the eleven explicit diagrams is a cross-interface coherence failure. Heavy splitting and flavor anisotropy are same-carrier perturbations when terminal support persists. A different microscopic source reproducing the full semantic trace is an observational mimic, not a unique-origin defeat. Protocol failure is not physical falsification. Parent proposals remain NotReady. Complete semantic-use agreement together with a contrary endpoint is a genuine counterexample burden.

8.3 Representative alternatives

Rival	Minimal difference locus	Disposition	Nonimplication
No-vectorlike reference	observable/record	structural separation; SPIC-conditional	no unique-origin claim
Law-only or pure hidden	anchor/source or terminal	different target/incomplete	no exclusion outside child

Rival	Minimal difference locus	Disposition	Nonimplication
Marginals agree, naturality fails	explicit coherence cell	CrossInterfaceCoherenceFailure	not a seventh classifier
Split heavy masses	perturbation	same carrier possible	not multiple carriers
Flavor anisotropy	perturbation	universality broken	rank/closure may survive
Rank loss	observable/source	rank boundary crossed	not automatically parent failure
$\Delta_{\text{act}} \neq \Omega_H$	observable/coherence	isometry/support relation fails	not detector-only under qualified SPIC
Local heavy excess only	record	background/different carrier	not CCST match
Same semantic trace, other source	microscopic origin	observational mimic	packet not unique-UV selector
Lost acceptance/control	protocol	ProtocolNotQualified	not physics falsifier
Parent proposal	target	ParentNotReady	child theorem unchanged
Complete semantic trace, contrary endpoint	all	GenuineCounterexample	reopens minimal false premise set

9 Carrier identity, continuum moduli, and exact unperturbed prediction

Theorem P8-12. One bearer per realization and continuum-moduli composition.

For every $b \in \mathcal{B}_8^{\text{ready}}$ and accepted $X \in \text{PhysMod}_{E, \text{VJ}}^b$,

$$|\text{TermCar}^b(X)| = 1, \quad |\pi_0 \text{SpecCore}^b(X)| = 4,$$

with three light sectors and one rank-three heavy sector. The bearer kind is `HybridLocalMatterCarrier`. Simultaneously,

$$|\pi_0 \text{PhysMod}_{E, \text{VJ}}^b| \geq \mathfrak{c}.$$

The common bearer kind does not identify the branch-specific minimal identity packets.

Proof.

PCR-6 proves that no proper sector, proper aggregate, or isolated higher relation has the complete terminal continuation signature. The source-connected parent C_X exists on one physical apex and contains all deletion-essential support. Source-support exhaustion shows that every independently eligible terminal-complete same-apex candidate is lawfully equivalent to that parent. A fresh within-realization ROC calculation gives one live bearer class and one assignment orbit. Separately, PCR-5E retains continuum many SCIC-distinct complete realizations across each child. These quantifiers do not conflict: the terminal quotient is computed inside each X , whereas the moduli quotient varies X . PCR-6 also proves that the heavy multiplicity descends to one rank-three sector under the complete lawful $U(3)_H$ action. The accidental and exact-global D2/D5 decorations remain distinct. PCR-8 merely composes these established levels and records their nonidentification [16, 17].

Thus

$$\text{OneBearerPerRealization}_{\text{VJ}} \oplus \text{ContinuumRealizationModuli} \oplus \text{TargetDependentMinimalIdentity}.$$

9.1 Exact source algebra of the common packet

The common unperturbed charged-current blocks are

$$N = c_E U, \quad V_H = -s_E U, \quad UU^\dagger = I_3.$$

Therefore

$$NN^\dagger = c_E^2 I_3, \quad V_H V_H^\dagger = s_E^2 I_3.$$

Since $c_E^2 + s_E^2 = 1$,

$$\Delta_{\text{act}} = I_3 - NN^\dagger = s_E^2 I_3 = \Omega_H.$$

The source-qualified child has $s_E > 0$, so Ω_H has rank three.

Theorem P8-13. Exact unperturbed cross-child observable prediction.

For every $b \in \mathcal{B}_8^{\text{ready}}$, every accepted $X \in \text{PhysMod}_{E,\text{VJ}}^b$, and every lawful child continuation preserving the frozen source,

$$\Delta_{\text{act}}(X) = \Omega_H(X) = \delta_E I_3, \quad C_H(X) = 0, \quad \text{rank } \Omega_H(X) = 3,$$

with structural heavy support $\text{Supp}_H^{\text{str}} = \text{Deg3}(H(I_E))$. In addition, for every licensed continuation u in the common-loop-kernel class

$$\mathcal{U}_{\text{loop}}^{\text{com}} := \{u : F_k(u) = F_H(u) \text{ for all three heavy sectors } k\},$$

the exact heavy off-diagonal CLFV amplitude vanishes. The packet is common to the two ready children; D2/D5 decorations remain child-indexed.

Proof.

The Gram identities follow from $N = c_E U$, $V_H = -s_E U$, $UU^\dagger = I_3$, and $c_E^2 + s_E^2 = 1$. Rank three follows from $\delta_E = s_E^2 > 0$. The structural support class follows from the degenerate rank-three source block on $H(I_E)$ and is not promoted to a physical unstable-state pole. For $u \in \mathcal{U}_{\text{loop}}^{\text{com}}$,

$$A_{\mu e}^H(u) \propto F_H(u) \sum_k U_{ek} U_{\mu k}^* = 0$$

by the common kernel and unitarity. The common-kernel premise is load-bearing: a continuation that resolves sector-dependent loop kernels leaves this exact CLFV-zero clause and is routed to the P8-15 symmetry/perturbation classification; it does not alter the Gram packet automatically.

PCR-7 proves naturality of the packet under lawful basis, gauge, matching, continuation, and child-groupoid morphisms. P8-02 and P8-06 identify it with the same accepted X and parent bearer. The no-vectorlike reference contains no active deficit or heavy structural support. Consequently the exact unperturbed packet is a child-qualified prediction lock, while the CLFV zero is stated exactly on the continuation class for which its imported proof applies.

$$\text{ExactChildPacket}_{\text{CCST}} \oplus \text{ExactUnperturbedPredictionLock}_{\text{CCST}} \oplus \text{CommonKernelCLFVZero}.$$

9.2 Minimality and nondecoupling

The minimal locked packet is relational:

$$O_{\text{CCST}}^{\text{min}} = (\Delta_{\text{act}}, \Omega_H, \text{Supp}_H^{\text{str}}).$$

The equality residual C_H and heavy CLFV zero are exact consequences but do not by themselves separate the reference. Removing Δ_{act} loses the light-side accounting relation. Removing Ω_H loses the inclusive heavy-side response. Removing $\text{Supp}_H^{\text{str}}$ allows a purely low-energy nonunitary response to mimic the continuous Gram coordinates. The only algebraic path to the no-vectorlike reference within the common source formula is $s_E \rightarrow 0$, which leaves the marked source-qualified child. The separation is therefore nondecoupling at the declared scope.

10 Nested structural robustness and symmetry-protected refinements

10.1 No single robustness radius

A structural radius is the distance, in a declared source/standing perturbation class, to the first failure locus of a specified feature. Different features use different traces and different metrics. The unperturbed theorem therefore has a feature vector rather than one confidence interval. Statistical confidence appears only after transport to an empirical record space.

10.2 Carrier trace cells

PCR-7.5 constructs finite source-support skeletons, partition decorations, exact deletion flags, and explicit faces on declared trace-transport cells. On each such cell the terminal signature and support data vary without crossing the promotion/source boundary. Define

$$\text{PhysMod}_{E, \text{VJ}}^{b, \text{cell-cert}} \subseteq \text{PhysMod}_{E, \text{VJ}}^b$$

to be the accepted realizations supplied with this cell certificate. For $X \in \text{PhysMod}_{E, \text{VJ}}^{b, \text{cell-cert}}$, let $U_{\text{car}}^b(X)$ be the connected component on which the support skeleton, terminal faces, deletion flags, and assignment orbit remain fixed.

10.3 Closure and rank domains

On the complete block-isometry class,

$$N' N'^{\dagger} + V'_H V_H'^{\dagger} = I_3,$$

so

$$\Delta'_{\text{act}} = I_3 - N' N'^{\dagger} = V'_H V_H'^{\dagger} = \Omega'_H.$$

This identity is exact throughout U_{iso} and does not require flavor universality or exact heavy degeneracy. For $V'_H = -s_E U + \delta V_H$ with $\eta_V = \|\delta V_H\|_2$, Weyl's singular-value inequality [27, 29, 31] gives

$$\sigma_{\min}(V'_H) \geq s_E - \eta_V.$$

Thus, if $\eta_V < s_E$,

$$\lambda_{\min}(\Omega'_H) = \sigma_{\min}(V'_H)^2 \geq (s_E - \eta_V)^2 > 0,$$

and rank $\Omega'_H = 3$. Aggregate heavy-projector stability additionally requires the full mass operator perturbation to remain below half the light-heavy spectral gap [28, 30].

Theorem P8-14. Nested structural-robustness composition.

The exact unperturbed one-carrier and CCST theorems hold for every accepted ready-child realization. Positive carrier radius is asserted only for $X \in \text{PhysMod}_{E, \text{VJ}}^{b, \text{cell-cert}}$. On the three feature domains:

1. $|\text{TermCar}^b(X_{\delta})| = 1$ on $U_{\text{car}}^b(X)$;
2. $\Delta'_{\text{act}} = \Omega'_H$ exactly on U_{iso} ;
3. rank $\Omega'_H = 3$ and $\lambda_{\min}(\Omega'_H) \geq (s_E - \eta_V)^2$ on U_{rank} .

The strongest joint statement holds only on

$$U_{\text{joint}}^b(X) = U_{\text{car}}^b(X) \cap U_{\text{iso}} \cap U_{\text{rank}}.$$

Uncertified accepted charts retain the exact unperturbed theorem and receive $\text{NotReady}_{\text{TraceTransportCell}}$ for a positive carrier radius.

Proof.

The carrier statement is imported from P75-13/P75-14, whose proof transports the finite support skeleton and exact terminal faces within a declared cell and then reapplies the PCR-6 terminal assignment orbit. It is not inferred from eigenvalue continuity. The closure statement is the block-isometry identity above. The rank statement follows from the singular-value lower bound and the separate heavy-projector spectral-gap condition. Since these proofs have different hypotheses, their conjunction is licensed only on their set-theoretic intersection. A universal cover of all accepted realizations by certified P75-13 cells is not in the handoff and is therefore listed as OB-12. This preserves the strongest exact result for all X while preventing a cell-local radius from being universalized.

$$\text{NestedStructuralRobustness}_{\text{VJ}}^{\text{certified cells}}.$$

10.4 Central and traceless heavy perturbations

Every Hermitian heavy mass-squared perturbation decomposes uniquely as

$$\delta K_H = aI_3 + T, \quad T = T^\dagger, \quad \text{tr } T = 0.$$

The commutant of the complete heavy $U(3)_H$ action consists of scalar operators. Therefore exact threefold degeneracy is preserved precisely on the central stratum $T = 0$. Every nonzero standing traceless component lifts the degeneracy to a $2 + 1$ or $1 + 1 + 1$ partition. If $\rho_K = \|T\|_2$, PCR-7.5 gives

$$\frac{3}{2}\rho_K \leq \text{diam Spec}(T) \leq 2\rho_K.$$

This is a spectral-sector theorem, not a terminal carrier-splitting theorem.

10.5 Universality and singular values

Flavor universality of the heavy inclusive Gram matrix means

$$\Omega'_H = \delta' I_3, \quad \delta' > 0.$$

For a full-rank V'_H , this holds exactly when the three singular values are equal. Rank three by itself permits arbitrarily small anisotropic perturbations and does not imply universality.

Theorem P8-15. Symmetry-protected refinement and zero radius.

Exact heavy threefold degeneracy is a $U(3)_H$ -protected class-exact feature on the central heavy-mass stratum. Exact flavor universality is class-exact on the equal-singular-value coupling stratum. Both have zero radius against unrestricted standing symmetry breaking. A nonzero traceless heavy perturbation may split the heavy spectral sector without changing terminal carrier cardinality inside $U_{\text{car}}^b(X)$.

Proof.

The commutant theorem gives the degeneracy statement. For every $\varepsilon > 0$, choose a nonzero traceless Hermitian T with $\|T\|_2 < \varepsilon$; it preserves the source dimension and may remain same-scope while lifting the degeneracy. Hence the unrestricted radius is zero. Similarly, choose a full-rank anisotropic perturbation δV_H with arbitrarily small norm whose singular values are unequal. Rank remains three for $\|\delta V_H\|_2 < s_E$, but $V'_H V'^{\dagger}_H$ is not scalar. Thus universality also has zero unrestricted radius. PCR-7.5 separately transports the terminal support trace on a certified carrier cell. Since a split eigenvalue does not supply a second complete terminal signature, spectral lifting does not imply carrier splitting. The exact implication is therefore restricted to

$$\text{Universality} \Rightarrow \text{Rank3}.$$

Degeneracy and universality are independent symmetry coordinates: neither implies the other. At the exact central PCR-7 source point, both happen to hold simultaneously because mass centrality and coupling isotropy are separately present in the frozen source packet; their joint truth there is a conjunction of source facts, not an implication. Meanwhile

$$\text{BlockIsometry} \Rightarrow \Delta'_{\text{act}} = \Omega'_H$$

and

$$\text{TerminalSupportComplete} \Rightarrow \text{OneCarrier}$$

are independent branches.

10.6 Summary of robustness domains

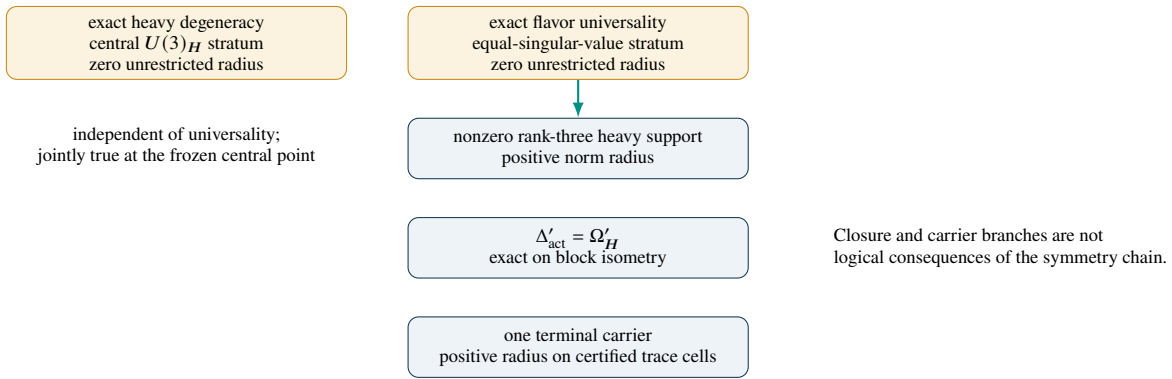


Figure 2: Feature-specific robustness. Only universality implies rank three; degeneracy is an independent symmetry coordinate.

Feature	Certified domain	Status	First failure
One carrier	$U_{\text{car}}^b(X)$ on declared P75-13 cells	positive radius	promotion/source/trace-cell boundary
Closure	U_{iso}	class-exact	loss or additional charged-current support
Rank three	$\eta_V < s_E$ plus heavy-projector gap	positive radius	singular-value/projector boundary
Heavy degeneracy	$T = 0$ central stratum	symmetry-protected exact	any nonzero standing traceless T
Flavor universality	equal singular values	symmetry-protected exact	anisotropic singular values
Joint robust core	$U_{\text{joint}}^b(X)$	intersection theorem	first failed component domain
Universal cell cover	all accepted X	NotReady	missing chart-cover theorem

11 Structural separation, SPIC, and a conditional testing framework

11.1 Three nonidentical gap objects

For matrices $A, B \in M_3(\mathbb{C})$ define the Frobenius product norm

$$\|(A, B)\|_{\text{pair}} := (\|A\|_F^2 + \|B\|_F^2)^{1/2}.$$

Let D_{pair} be a compact structural domain satisfying

$$D_{\text{pair}} \subseteq U_{\text{iso}} \cap U_{\text{rank}}, \quad \eta_V \leq \bar{\eta}_V < s_E.$$

The ideal pair-space gap from the zero no-vectorlike pair is

$$g_{\text{pair}}^{\text{str}} := \inf_{z \in D_{\text{pair}}} \|(\Delta'_{\text{act}}(z), \Omega'_H(z)) - (0, 0)\|_{\text{pair}}.$$

Block isometry gives $\Delta'_{\text{act}} = \Omega'_H$, while rank preservation gives $\lambda_{\min}(\Omega'_H) \geq (s_E - \bar{\eta}_V)^2$. Hence

$$g_{\text{pair}}^{\text{str}} \geq \sqrt{6} (s_E - \bar{\eta}_V)^2 > 0.$$

This is not a mass gap. On a separately declared mass-support chart, $g_{\text{mass}}^{\text{str}}$ denotes the true structural distance between the light and heavy support classes. The reconstructed observed interval gap is $\widehat{g}_{\text{mass}}^{\text{str}}$. Neither object is assigned a numerical VJ pole value here.

11.2 The SPIC object

A SPIC supplies carrier and reference response maps into one record space; signal, control, sideband, and nuisance models; covariance and coarsening rules; sampling and coverage; deterministic floors; attainable record radius; same-record separator; inverse-radius, mass-support, rank, and identifiability guards; and held-out information ancestry. It is a detector/record certificate, not a new physical source premise.

Theorem P8-16. Structural pair-space separation and SPIC-conditional record selection.

On $D_{\text{pair}} \subseteq U_{\text{iso}} \cap U_{\text{rank}}$ with $\eta_V \leq \bar{\eta}_V < s_E$,

$$g_{\text{pair}}^{\text{str}} \geq \sqrt{6} (s_E - \bar{\eta}_V)^2 > 0.$$

Detector-level same-record selection is licensed only after an instantiated SPIC supplies

$$\begin{aligned} s_{\text{car,ref}} &> 0, & 2r_{\text{rec}} &< s_{\text{car,ref}}, \\ \frac{2r_{\text{rec}}}{\kappa_{\text{car}}} &< r_{Z,\text{car}}, & \frac{2r_{\text{rec}}}{\kappa_{\text{ref}}} &< r_{Z,\text{ref}}, \end{aligned}$$

together with nuisance, coarsening, floor, coverage, Gram/rank, and identifiability guards. If the protocol claims mass-resolved support, the separate conditions are

$$\begin{aligned} r_m &< \frac{g_{\text{mass}}^{\text{str}}}{4} && \text{for uniform design,} \\ g_{\text{mass}}^{\text{str}} &> 2r_m && \text{for true-support matching,} \\ \widehat{g}_{\text{mass}}^{\text{str}} &> 2r_m && \text{for observed interval resolution.} \end{aligned}$$

No equality among $g_{\text{pair}}^{\text{str}}$, $g_{\text{mass}}^{\text{str}}$, $\widehat{g}_{\text{mass}}^{\text{str}}$, $s_{\text{car,ref}}$, r_{rec} is licensed. The present detector status is conditional because PCR-8 does not instantiate a complete experiment-specific SPIC.

Proof.

On U_{iso} , the complete charged-current block isometry gives $\Delta'_{\text{act}} = \Omega'_H$. On U_{rank} , singular-value control gives $\lambda_{\min}(\Omega'_H) \geq (s_E - \bar{\eta}_V)^2$. A positive semidefinite 3×3 matrix with this minimum eigenvalue has Frobenius norm at least $\sqrt{3}(s_E - \bar{\eta}_V)^2$. The two equal pair coordinates therefore give the $\sqrt{6}$ bound. Without block isometry, rank positivity alone does not supply the second equal coordinate, so the intersection hypothesis is load-bearing. SPIC then forms carrier and reference response classes in one record quotient. The half-separator, inverse-radius, and response/nuisance/coverage conditions imply disjoint confidence inverse regions. Mass-support resolution is a different question and uses $g_{\text{mass}}^{\text{str}}$ or $\widehat{g}_{\text{mass}}^{\text{str}}$. A transport inequality $s_{\text{car,ref}} \geq \kappa_{\text{sep}} g_{\text{pair}}^{\text{str}}$ may be used only when SPIC certifies the response map, domain, κ_{sep} , and nuisance conditions. PCR-8 contains no unconditional identification of structural and record geometry.

$$\text{StructuralPairGap}_{\text{CCST}} \oplus \text{ConditionalRobustPredictionLock}_{\text{CCST|SPIC}}.$$

11.3 Finite exposure and current experimental status

PCR-7 proves existence in principle of a finite protocol and finite exposure on the frozen compact source-qualified domain. PCR-8 does not assert that an existing trigger, reconstruction, or published analysis realizes the full SPIC. Current implementation requires experiment-specific efficiencies, mass/vertex resolution, backgrounds, covariance, control regions, systematic correlations, and record inverse bounds:

$$\text{CurrentFacilitySPIC} = \text{NotReady}.$$

11.4 Prospective test design

The test is flavor-complete charged-current spectral tomography: reconstruct light normalization and appearance; scan inclusive heavy support; reconstruct Δ_{act} and Ω_H under common controls; retain support, rank, and closure in one record quotient; and

compare the frozen carrier/reference families under an instantiated SPIC. A local heavy excess without the correlated active deficit is not the packet; a deficit without inclusive heavy weight indicates another source or loss channel.

11.5 Selection does not imply unique microscopic origin

SPIC distinguishes the frozen carrier class from the frozen no-vectorlike reference. It does not exclude every richer source capable of the same complete packet. Such a source is an observational mimic and requires a separate source-selection target.

12 Search benchmark, weak normalization, and complex-pole scope

Theorem P8-17. Authority of the source-based search benchmark.

The pair

$$B_{D3} = ([28, 30] \text{ GeV}, c\tau_A^{\text{resp}} = 0.2 \text{ mm})$$

has status

SourceCertified_ATLASReweightGridAnchored_SearchBenchmark.

Only 30 GeV is a direct ATLAS mass anchor; 0.2 mm is an exact member of the discrete lifetime-reweight grid; and the HEPData ordinate replay is a simplified-model compatibility screen rather than a source-complete recast. The benchmark does not define $H(I_E)$ and is not a VJ pole-mass or proper-lifetime prediction.

Proof.

PCR-5D3 constructs a source-first continuum on $[28, 30]$ GeV with positive direct-state, active-closure, and CLFV margins in its fixed response specialization [12, 13]. PCR-7.5 retains this as an independently source-certified scouting window and adds the experimental authority chain: direct 30 GeV anchor, exact discrete 0.2 mm grid coordinate, conditional ordinate screen, and on-shell- W kinematics [19, 35, 36, 37, 38, 34]. None of these objects supplies the full X7/X8 physical pole or lifetime. The 28 GeV endpoint is source-theoretic, not directly simulated in the cited grid; the response coordinate is not a full-source proper lifetime; and a pp laboratory displacement requires boost, angular, acceptance, and reconstruction convolution. Therefore the pair is scientifically motivated without becoming a numerical prediction.

12.1 Trigger and selection architecture

The benchmark motivates an initial experiment map rather than a theorem of present sensitivity:

- flavor-complete charged-current parent tagging;
- LNC/LNV-inclusive heavy reconstruction so that exact-global and accidental children are not silently selected by the trigger;
- prompt and near-prompt categories around the discrete response grid point;
- inclusive heavy-support reconstruction rather than a single exclusive decay assumption;
- common active-normalization, sideband, and control records needed for $\Delta_{\text{act}}\text{--}\Omega_H$ closure;
- resolution-adaptive refinement only after a frozen benchmark analysis is defined.

This strategy answers why the band is worth scouting; it does not claim a source-complete recast of existing analyses.

Theorem P8-18. Marked-child normalization and weak-record status.

The published marked-child source bound remains

$$0 < \epsilon_{\text{row}} \leq \frac{1}{256}.$$

The frozen HFLAV reconstruction has status ConditionalLightResponseCompatibleAt95 with $p = 0.1349$ and no common- ϵ information. The separate leading-response W -mass calculation has status ScreeningOnly_NotCertifiedForX7

with

$$\epsilon_{95}^{\text{scr}} \simeq 1.62 \times 10^{-3}.$$

The jointly covered physical endpoint remains

$$\epsilon_{95}^{\text{weak}} = \text{NotReady}.$$

Proof.

The one-pass rule was frozen before numerical disposition:

$$\epsilon_{95}^{\text{weak}} < \frac{1}{256} \Rightarrow \text{publish tightened source-weak intersection; otherwise retain } \frac{1}{256} \text{ and report compatibility only.}$$

The HFLAV ratio block contains an exact algebraic redundancy and is evaluated on the independent basis (A, C, D, E) [39, 19]. Its covariance result is compatible with the marked child but carries zero information on a common ϵ . It also shares LFU ancestry with the D3 child formation and therefore is not independent validation. The W -mass screen [40, 34] has numerical leverage but lacks a jointly certified finite-mass HNL response, open-channel and loop treatment, electroweak input scheme, template/acceptance model, theory covariance, and uniform source-domain coverage. It cannot be renamed $\epsilon_{95}^{\text{weak}}$. Since the covered endpoint is NotReady, the predeclared stop rule returns the original $1/256$ bound. Reopening the pass merely to obtain a smaller number would violate the empirical-use and anti-bias firewalls.

12.2 Requirements for a future weak constraint

A future fixed-child conditional normalization constraint requires an independent weak-rate record and a declared electroweak input-scheme response that constrains ϵ_{row} conditional on the child. It must be independent of the D3 prompt-lifetime calibration and of the later lifetime/response record used for mass inversion. A stronger child-selection theorem requires a record acting across a comparison class not already restricted to the marked child. Neither the dyadic rule nor its upper bound performs that work.

Theorem P8-19. Conditions for a physical complex-pole interpretation.

Structural heavy support is exact as $\text{Supp}_H^{\text{str}}$ on the symbolic domain $H(I_E)$. A physical unstable-state pole is a separate conditional object

$$\text{Pole}_H^{\text{phys}} = \{z : \det D^{-1}(z) = 0\}$$

with sheet, continuation, noncommuting width/mass, and coherent-response data. Numerical physical mass and lifetime remain NotReady until the complete X7/X8 certificates are supplied.

Proof.

The PCR-6/PCR-7 source mass block and structural support do not by themselves define the analytic continuation of an unstable state. X7 requires the complete inverse propagator, self-energy and width kernel, sheet/continuation prescription, source-qualified normalization intersection, and treatment of noncommuting mass and width matrices. Width eigenvalues suffice only in a commuting, controlled-energy-dependence reduction; otherwise the physical object is the unordered complex-pole set together with coherent response. X8 additionally requires an independent response record, a coverage-preserving union of polewise-identifiable and coherent-response projections, exhaustive compatible response types, a joint nuisance envelope, and the mass/input-scheme bridge. The response-chart image $\beta\gamma c\tau_A^{\text{resp}}$ is not a proper lifetime or a pp laboratory mean. Because those gates are not supplied, PCR-8 closes only the authority theorem and conditional schema.

$$\text{ConditionalComplexPoleEnvelope}_{H(I_E)} \oplus \text{X7X8InputsNotReady}.$$

12.3 Summary of numerical scope

Object	Status	Authorized statement	Blocked statement
$H(I_E)$	structural exact	symbolic heavy domain	numerical interval
B_{D3}	search benchmark	initial scouting architecture	predicted pole mass/lifetime
ϵ_{row}	target-qualified bound	$0 < \epsilon_{\text{row}} \leq 1/256$ in marked child	independent weak measurement
HFLAV	compatibility check	child-compatible at 95%	independent validation/normalization
$\epsilon_{95}^{\text{scr}}$	screen	numerical screening leverage	covered physical endpoint
$\epsilon_{95}^{\text{weak}}$	NotReady	named future burden	substituted screen
$\text{Supp}_H^{\text{str}}$	structural support	rank/multiplicity/support class	unstable-state pole
$\text{Pole}_H^{\text{phys}}$	conditional	X7/X8 complex-pole schema	current numerical mass/lifetime

13 Atomic feature claims and held-out evaluation

13.1 Atomic claims and logical dependencies

The falsifier lattice is finer than the publication theorem numbering. Let G_g^{atom} be the finite DAG whose nodes are independently falsifiable claim coordinates. Each node records its exact source theorem, target/feature domain, empirical applicability conditions, direct predecessors, preserved nonimplications, permitted update route, and containing P8 theorem. The principal nodes are

ReadyChildScope, OneApex, OneBearer, CarrierRadius, NonzeroHeavySupport, Rank3, Closure, Universality, Deg3, ReferenceComplete, PairGap, SPICQualified, RecordSeparation, CrossChildCore, BenchmarkAuthority, WeakBound, PoleConditional, HeldOutEligible.

The graph explicitly omits false implications such as $\text{Deg3} \Rightarrow \text{OneBearer}$ and $\text{Rank3} \Rightarrow \text{Universality}$.

For a qualified record R , let

$$\text{Fail}_{\text{atom}}(R) = \{a \in V(G_g^{\text{atom}}) : R \text{ contradicts the applicable atomic claim } a\}.$$

Then

$$\text{MinFail}_{\text{atom}}(R) = \{a \in \text{Fail}_{\text{atom}}(R) : \nexists a' \in \text{Fail}_{\text{atom}}(R), a' \prec_{\text{atom}} a\}.$$

The theorem-reporting map $\kappa(a)$ returns the containing P8 theorem only after the atomic antichain is fixed.

Theorem P8-20. Localization by minimal failure antichains.

Every predeclared qualified deviation is mapped to $\text{MinFail}_{\text{atom}}(R)$, its theorem-reporting image $\kappa(\text{MinFail}_{\text{atom}}(R))$, its admissible update routes, and the exact nonimplications that remain. If SPIC, independence, freeze, or coverage fails, the record first receives ProtocolNotQualified and no physical failure set is inferred. Parent records are outside the ready-child atomic poset.

Proof.

The atomic claim DAG is formed before any observed record. Its edges are logical dependence edges, not visual ordering. Every falsifier-table identifier corresponds to an atomic node whose domain matches its source theorem, and the resulting graph is acyclic. On qualified data the evaluator forms the contradicted atomic set and removes every node with a contradicted predecessor. The remainder is an antichain and need not be singleton. The reporting map then translates atomic nodes to P8 theorem labels without adding causal priority.

Resolved heavy splitting with rank and closure intact gives {Deg3}; anisotropic weights with rank and closure intact give {Universality}; independent qualified failures may give incomparable nodes. A protocol defect is reported separately. Response directions outside the stated model receive NewResponseDirection rather than being forced into an existing node.

$$\text{AtomicClaimDAGClosed}_8 \oplus \text{MinimalFailureAntichainClosed}_8.$$

13.2 Feature-level falsifier matrix

Observation with stated qualifiers	Atomic minimal claim set	Preserved nonimplications
Heavy splitting; packet/rank/closure qualified	{AC-DEG3}	one carrier, rank, closure may survive
Unequal flavor weights; rank/closure qualified	{AC-UNIVERSALITY}	no automatic bearer defeat
Rank below three on U_{rank} $\Delta_{\text{act}} \neq \Omega_H$ on U_{iso}	{AC-RANK3} {AC-CLOSURE}	no parent closure or new carrier selection support/isometry/source issue remains to localize
No heavy support with complete coverage	{AC-NONZERO-SUPPORT}	unqualified detector null is insufficient
Second complete bearer; terminal interface fixed	{AC-ONE-BEARER}	spectral splitting alone is insufficient
Carrier/reference overlap under complete SPIC	{AC-RECORD-SEPARATION}	ideal pair gap not automatically false
Child packets differ after common-source qualification	{AC-CROSS-CHILD-CORE}	reopen common packet clause
Protocol not qualified	no physical failure set	report protocol status only
No benchmark signal	empty absent a sensitivity theorem	benchmark is not discovery guarantee
Parent discrepancy	outside child failure poset	child theorem remains intact

Theorem P8-21. Held-out outcomes and post-test reporting.

The prediction is frozen before qualified empirical comparison. A future data set yields a PCR-8 evaluation only if it passes use-level information independence, protocol identity, nuisance freeze, SPIC, coverage, and no-retuning conditions. The current status is

$$\text{FrozenPredictionAwaitingQualifiedEvaluation}.$$

Proof.

PCR-7 froze the ready-child tags, packet, protocol, reference family, confidence relation, and falsifier before held-out records. P8-08 refines empirical ancestry, P8-16 requires SPIC for detector selection, and P8-20 requires atomic antichain reporting. A qualifying evaluation must match $p_\star$ or a precertified refinement, use information absent from formation ancestry, freeze nuisance and covariance before unblinding, and report preregistered inverse regions and the complete atomic minimal-failure set without retuning. No such completed record is imported here. This theorem establishes readiness and outcome typing, not confirmation.

13.3 Permitted held-out outcomes

Outcome	Condition	Interpretation
CarrierPacketMatched	carrier inverse contains truth; reference inverse empty	target-qualified support, not unique origin
ReferencePacketMatched	reference inverse contains truth; carrier inverse empty	ready-child packet falsified or updated
PartialFeatureFailure	one or more qualified coordinates fail	report complete $\text{MinFail}_{\text{atom}}(R)$
BothInverseRegionsEmpty	neither frozen family covers the record	new response/source or protocol defect
ProtocolNotQualified	SPIC, coverage, independence, or freeze fails	no empirical physics verdict
InconclusiveAtResolution	separator or guards not achieved	prediction remains frozen

13.4 No retrospective rescue

A post-unblinding change to packet, source domain, reference, mass window, nuisance model, control set, or falsifier creates a new prediction version. It may be scientifically useful, but it is not the prelocked PCR-8 evaluation.

14 Open boundaries and the child-qualified capstone theorem

Theorem P8-22. Open boundaries of the child-qualified result.

The following remain explicitly open or conditional: the unresolved parent neutrino family; THUCL-D3-1.1 global completion; the complete D4 likelihood; detector-specific SPIC and held-out evaluation; independent weak normalization $\epsilon_{95}^{\text{weak}}$; numerical $\text{Pole}_H^{\text{phys}}$ mass/lifetime; present-facility sensitivity; universal trace-cell coverage; unique microscopic origin; cross-fermion universality; and complete historical influence certification. No open item is used to prove an exact positive clause.

Proof.

Each open burden has a target, status, reopening requirement, and preserved endpoint. Missing SPIC does not weaken the exact pair-space gap; missing trace-cell coverage does not weaken the universal unperturbed one-bearer theorem; missing X7/X8 does not weaken structural support; and the unresolved parent does not weaken the two child theorems. The typed handoff pullback, reconciled dependency structure, quotient diagram, endpoint-erasure argument, and capstone trace close the local PCR-8 composition requirements without filling any of these external frontiers.

OpenEndpointClosure₈.

Theorem P8-23. Child-qualified prediction closure.

Let $b \in \mathcal{B}_8^{\text{ready}}$ and $X \in \text{PhysMod}_{E,VJ}^b$. Then:

1. The three final handoffs form the typed homotopy pullback $\mathcal{H}_8^b$ over $\mathcal{I}_b$, with their section functors bound to P6-26, P7-28, and P75-26. The scientific dependency graph is reconciled with the uses made in the P8 proofs, while ambient mathematical authorities are treated separately.
2. Exact upstream source signatures reconcile with the independently formed endpoint-erasure inputs. No later carrier, packet, robustness, benchmark, pole, or empirical-outcome field enters an earlier formation signature.
3. Each accepted X has exactly one terminal carrier that bears the licensed continuation. Its common source-qualified kind is `HybridLocalMatterCarrier`; its standing core has three nondegenerate light sectors and one rank-three heavy sector. The realization quotient remains continuum-sized and the child decorations remain distinct.
4. At the exact unperturbed source,

$$\Delta_{\text{act}} = \Omega_H = \delta_E I_3, \quad \text{rank } \Omega_H = 3, \quad \text{Supp}_H^{\text{str}} = \text{Deg3}(H(I_E)).$$

The heavy CLFV zero holds on the licensed common-loop-kernel continuation class.

5. If X lies on a P75-13-certified trace cell, one carrier persists on $U_{\text{car}}^b(X)$. On U_{iso} , $\Delta'_{\text{act}} = \Omega'_H$ exactly. On U_{rank} , heavy support remains nonzero and rank three. Joint robustness is asserted only on $U_{\text{joint}}^b(X)$.
6. Exact threefold heavy degeneracy and exact flavor universality are stronger symmetry-protected refinements with zero unrestricted standing-breaking radius.
7. On $D_{\text{pair}} \subseteq U_{\text{iso}} \cap U_{\text{rank}}$, the exact ideal separation is $g_{\text{pair}}^{\text{str}} > 0$. Same-record selection is conditional on SPIC; mass-support and observed-interval resolution use $g_{\text{mass}}^{\text{str}}$ and $\hat{g}_{\text{mass}}^{\text{str}}$.
8. The D3 28–30 GeV, 0.2 mm pair remains a search benchmark; $0 < \epsilon_{\text{row}} \leq 1/256$ is a marked-child source bound; $\epsilon_{95}^{\text{weak}}$, numerical $\text{Pole}_H^{\text{phys}}$ mass/lifetime, current sensitivity, universal trace-cell coverage, and parent closure remain NotReady.
9. Every finitely presented same-scope capstone use descends to the explicit $\mathcal{K}_{\text{cap}}$ diagram; semantic use equality derives capstone trace identity and rival disposition. Every qualified future discrepancy is reported on the separately frozen atomic claim DAG by its minimal-failure antichain.

Consequently,

$$\begin{aligned} \text{Status}_{\text{PCR8}} = & \text{ChildQualifiedDependencyCleanPredictionClosure}_{\mathcal{B}_8^{\text{ready}} \text{ VJ}} \\ & \oplus \text{ExactUnperturbedPredictionLock}_{\text{CCST}} \\ & \oplus \text{NestedStructuralRobustness}_{\text{certified cells VJ}} \\ & \oplus \text{StructuralPairGap}_{U_{\text{iso}} \cap U_{\text{rank}}} \\ & \oplus \text{ConditionalRobustPredictionLock}_{\text{CCST|SPIC}} \\ & \oplus \text{FrozenPredictionAwaitingQualifiedEvaluation}. \end{aligned}$$

Proof.

P8-02 supplies the bound handoff pullback. P8-03 compares the scientific dependency graph with the independently reviewed proof-use structure and the ambient mathematical authorities. P8-05/P8-06 supply quotient and one-apex coherence. P8-07 reconciles exact source theorem signatures before forming endpoint-erasure triangles. P8-08–P8-10 establish empirical, numerical, and evidentiary authority. P8-11A derives trace identity from the independently frozen semantic capstone-use grammar and explicit diagram equations; P8-11B dispositions rivals. P8-12–P8-19 supply the imported physical, robustness, gap, benchmark, weak, and pole-authority conclusions at their exact domains. P8-20 constructs the atomic feature DAG used by P8-21, and P8-22 closes every named open boundary. Thus the dependency-clean conjunct is supported by four independent reconciliations—proof use, source signature, semantic capstone use, and atomic falsifier structure—rather than by self-consistency of any single representation.

14.1 Summary of conclusions

Question	PCR-8 endpoint	Exact interpretation
Handoff composition	typed homotopy pullback	explicit common groupoid, functors, cells, universal property
Dependency closure	proof-use structure reconciled	scientific uses agree; ambient mathematics separate
Quotient closure	exact structural descent	record leg remains SPIC-conditional
Rival closure	semantic-use complete capstone trace	explicit diagram objects, arrows, and commuting equations
Terminal bearer	one per accepted realization	within-realization theorem
Cross-realization quotient	continuum	no unique realization selection
Unperturbed packet	exact	child-qualified and structural-support normalized
Carrier robustness	positive on certified cells	missing charts NotReady for radius only
Closure/rank	intersection-typed	joint only on $U_{\text{joint}}^b(X)$
Degeneracy/universality	symmetry-protected	zero unrestricted radius
Pair-space gap	exact on $U_{\text{iso}} \cap U_{\text{rank}}$	not mass or confidence gap
Record selection	SPIC-conditional	distinct separator/radius/guard objects
Falsifier localization	atomic minimal antichain	every feature label is an atomic claim in the stated graph
Search window	benchmark	not mass/lifetime prediction
Weak normalization	retain 1/256	physical weak endpoint NotReady
Physical pole	X7/X8 conditional	numerical mass/lifetime NotReady
Parent/all-neutrino theorem	NotReady	no child-to-parent transport

15 Physical interpretation

15.1 What the carrier theorem now says

Within each completed vectorlike-joint child realization, the neutrino architecture is not six carriers corresponding to six displayed eigenvectors, nor three carriers corresponding to three light masses. It is one source-connected continuation-bearing carrier with a standing internal spectrum. Three light sectors remain nondegenerate and boundary-visible; the heavy multiplicity is one rank-three standing sector under the lawful $U(3)_H$ action. The carrier contains the internal relations needed to realize oscillation, active closure, matching, population, symmetry, and branch-specific conjugation structure on one apex.

This is compatible with continuum many complete realizations. Carrier identity asks what is the terminal bearer inside one accepted realization. The moduli quotient asks how many complete source/physical realizations remain across the target. The answer can be one to the first and continuum to the second without contradiction.

15.2 What the observable theorem now says

The most informative prediction is relational. A source-qualified active deficit is accompanied by inclusive heavy charged-current support with equal total weight at the unperturbed point:

$$\Delta_{\text{act}} = \Omega_H.$$

The heavy support is rank three and structurally present in the source-qualified domain. A local heavy excess alone is not the packet. A deficit alone is not the packet. The relation between them, together with structural support, is what separates the ready-child carrier class from the no-vectorlike reference.

15.3 What remains robust when symmetry is relaxed

The deepest robust content is not exact degeneracy. Small standing traceless mass perturbations split the heavy spectrum immediately. Small anisotropic coupling perturbations spoil exact flavor universality while preserving rank. What persists on the certified joint domain is one terminal carrier, nonzero rank-three heavy support, and exact light-heavy charged-current

accounting. This is a stronger scientific statement than advertising a fragile exact triple pole, because it identifies the invariant core and the precise experiments that would distinguish a refinement failure from a carrier failure.

15.4 Why the benchmark is useful without being a prediction

The 28–30 GeV, 0.2 mm benchmark is not arbitrary: it lies in a source-certified D3 continuum with positive margins, touches a direct 30 GeV response anchor, uses an exact lifetime reweight-grid point, and sits in a kinematic regime suited to tagged charged-current parent events and prompt/near-prompt reconstruction. Those facts justify allocating experimental attention. They do not derive the VJ physical pole mass or lifetime, and they do not promise a signal at current sensitivity. The distinction between “motivated place to look” and “predicted value” is part of the theorem’s evidential integrity.

15.5 Why a confidence interval is not attached to the structural theorem

The carrier and packet conclusions are exact conditional theorems. Their deterministic robustness is expressed by feature-specific source/standing margins. Frequentist or Bayesian confidence applies only to an empirical implementation of the protocol and its record-space inverse. Converting a structural radius into a statistical confidence interval would conflate two mathematical spaces. SPIC is the bridge that keeps them related but distinct.

15.6 What a future result would mean

A resolved heavy split would test the exact $U(3)_H$ symmetry refinement, not necessarily the one-carrier theorem. Unequal flavor weights would test the equal-singular-value stratum. Rank loss would test the source/coupling margin. A failure of $\Delta_{\text{act}} = \Omega_H$ under complete qualified coverage would indicate missing or additional charged-current support and hence a deeper source or target revision. A second independently complete terminal bearer would directly attack the PCR-6 source-support exhaustion and ROC. A correct packet in a different microscopic theory would show observational mimicry, not failure of the packet. This localization makes the programme scientifically revisable without making it unfalsifiable.

16 Conclusion

For the two completed source-qualified vectorlike-joint sectors, the imported carrier and observable results compose on a single accepted physical apex. Every accepted realization has one terminal continuation-bearing hybrid local matter carrier, whose standing spectrum contains three nondegenerate light sectors and one rank-three heavy sector. The quotient of accepted realizations remains continuum-sized, so this within-realization uniqueness does not select one model across the full moduli space.

At the unperturbed source, the charged-current projection satisfies

$$\Delta_{\text{act}} = \Omega_H = \delta_E I_3, \quad \text{rank } \Omega_H = 3.$$

Structural heavy support is retained in the symbolic domain $H(I_E)$, and the common-loop-kernel continuation class gives the stated vanishing heavy CLFV amplitude. These are child-qualified structural results, not a numerical heavy-particle mass or lifetime prediction.

Robustness depends on the feature under consideration. Carrier identity has positive radius on certified trace-transport cells; the closure identity is exact on the block-isometry class; rank three has a positive norm radius; and exact degeneracy and flavor universality are independent symmetry-protected properties with zero unrestricted symmetry-breaking radius. The strongest joint statement is confined to $U_{\text{joint}}^b(X)$.

The ideal carrier/reference separation is likewise structural. Detector-level discrimination requires an instantiated SPIC with a common record space, response and nuisance models, coverage, resolution, and identifiability bounds. The 28–30 GeV, 0.2 mm pair is a search benchmark rather than a pole prediction, and the marked-child bound $0 < \epsilon_{\text{row}} \leq 1/256$ does not supply the unresolved physical weak endpoint. Qualified held-out evaluation, present-facility sensitivity, a numerical complex pole, the complete D4 likelihood, universal trace-cell coverage, the parent family, unique microscopic origin, and cross-fermion extensions remain open.

The resulting status is

$$\begin{aligned} \text{Status}_{\text{PCR8}} = & \text{ChildQualifiedDependencyCleanPredictionClosure}_{\mathcal{B}_8^{\text{ready}} \text{VJ}} \\ & \oplus \text{ExactUnperturbedPredictionLock}_{\text{CCST}} \\ & \oplus \text{NestedStructuralRobustness}_{\text{VJ}}^{\text{certified cells}} \\ & \oplus \text{ConditionalRobustPredictionLock}_{\text{CCST|SPIC}} \\ & \oplus \text{FrozenPredictionAwaitingQualifiedEvaluation}. \end{aligned}$$

The semantic trace classifies finitely presented alternatives only within the declared same-scope use grammar. Accordingly, the paper establishes a child-qualified composition and prediction theorem together with a conditional falsification framework, while leaving all broader physical and empirical questions explicit.

A Notation and status calculus

Symbol	Meaning
$\mathcal{B}_8^{\text{ready}}$	the two completed vectorlike-joint children $b_{\text{acc}}, b_{\text{glob}}$
$\mathcal{A}_b$	accepted ready-child physical-apex groupoid
$\mathcal{I}_b$	common cross-handoff identity groupoid
$\mathcal{H}_8^b$	typed three-handoff homotopy pullback
$X \in \text{PhysMod}_{E, \text{VJ}}^b$	accepted source-faithful one-apex realization in child b
C_X	source-connected terminal bearer in X
$\text{TermCar}^b(X)$	within-realization terminal carrier quotient
N, V_H	light and heavy charged-current blocks
Δ_{act}	active charged-current deficit $I_3 - NN^\dagger$
Ω_H	inclusive heavy Gram matrix $V_H V_H^\dagger$
$H(I_E)$	symbolic source-qualified heavy domain
$\text{Supp}_H^{\text{str}}$	structural heavy support/multiplicity object
$\text{Pole}_H^{\text{phys}}$	physical unstable-state complex-pole object, conditional on X7/X8
$U_{\text{car}}^b(X)$	carrier-stability component on a certified trace cell
U_{iso}	complete charged-current block-isometry class
U_{rank}	rank and heavy-projector stability domain
$U_{\text{joint}}^b(X)$	$U_{\text{car}}^b(X) \cap U_{\text{iso}} \cap U_{\text{rank}}$
$g_{\text{pair}}^{\text{str}}$	ideal pair-space structural gap on $U_{\text{iso}} \cap U_{\text{rank}}$
$g_{\text{mass}}^{\text{str}}$	true structural light-heavy support gap on a declared mass chart
$\widehat{g}_{\text{mass}}^{\text{str}}$	reconstructed observed support-interval gap
$s_{\text{car,ref}}$	SPIC same-record carrier/reference separator
r_{rec}	attainable common-record confidence radius
B_{D3}	([28, 30] GeV, 0.2 mm) search benchmark
ϵ_{row}	marked-child row deficit, $0 < \epsilon_{\text{row}} \leq 1/256$
$\epsilon_{95}^{\text{scr}}$	screening-only leading-response endpoint
$\epsilon_{95}^{\text{weak}}$	jointly covered physical weak endpoint, currently NotReady
$\text{MinFail}(R)$	minimal failed antichain in the qualified claim poset

A.1 Status logic

Undefined/NotReady means that a named typing, source, target, continuation, protocol, response, or provenance burden is missing. It is not a third truth value. On the evaluated fragment, standing and theorem claims remain bivalent. A NotReady

object may become defined when the missing scientific conditions are supplied; until then it is neither false nor available as a premise.

B Source authority and dependency structure

B.1 Direct scientific sources

ID	Source	Version	PCR-8 use
AUTH-P6	<i>Physical Nature and Minimal Identity of the Neutrino Carrier</i>	final handoff	terminal bearer and spectral core
AUTH-P7	<i>Observable Projection and Experimental Prediction Lock</i>	1.3.0	packet, protocol, reference, held-out lock
AUTH-P75	<i>Structural Stability and Degeneracy-Lifting</i>	1.4.4	robustness, SPIC, benchmark, weak/pole authority
AUTH-ARC	<i>Constraint-Induced Physical Carrier Realization</i>	Aug. 2026	PCR-8 mandate
AUTH-SKEL	<i>PCR-8 Detailed Theorem Skeleton</i>	0.2.0	theorem specification

B.2 Three dependency representations

Representation	Definition	Role
$G_{8,sci}^{reg}$	scientific theorem/object formation map	intended direct scientific dependencies
$G_{8,sci}^{use}$	theorem-local uses identified independently in the P8 proofs	scientific uses in the arguments
G_8^{math}	separate ambient mathematical authority structure	categorical, matrix, finite-poset, quotient, and confidence tools

For every P8 theorem, the scientific formation and proof-use structures agree, and the resulting graph is acyclic. Upstream source formation is examined separately in P8-07. This dependency analysis records the use of the upstream theorems; it does not claim to re-prove them.

B.3 Representative late-stage dependencies

From	To	Exported object	Blocked stronger reading
PCR-5E children	PCR-6/P8-01	ready child groupoids	no parent universalization
PCR-6 bearer	PCR-7/P8-12	terminal interface and one bearer	no detector result selects carrier
PCR-7 packet	PCR-7.5/P8-13	exact packet and protocol	no future record forms packet
PCR-7.5 handoff	P8-02/P8-14–19	robustness and authority	no generic symmetry robustness
P8-02 pullback	P8-05/06/11A	coherent handoff cone	no identity from notation alone
P8-03 reconciliation	P8-07/20/23	verified premise order	independent proof-use comparison required
P8-11A trace	P8-11B/20/23	semantic-use-derived object/coherence trace	no definition-by-trace circularity

From	To	Exported object	Blocked stronger reading
P8-16 gaps	P8-20/23	pair gap and conditional record test	no mass/confidence conflation

C Object crosswalk, handoff groupoid, and quotient diagrams

C.1 Stable object identities

ID	Symbol	Scientific object	Blocked conflation
OBJ-REAL	X	accepted ready-child realization	not unique selected model
OBJ-BEARER	C_X	terminal bearer within X	not a spectral sector
OBJ-DELTA	Δ_{act}	active-deficit Gram coordinate	not weak-rate scalar
OBJ-OMEGA	Ω_H	inclusive heavy Gram coordinate	not one exclusive channel
OBJ-HSUPP	$\text{Supp}_H^{\text{str}}$	structural heavy support	not physical complex pole
OBJ-HPOLE	$\text{Pole}_H^{\text{phys}}$	X7/X8 unstable-state pole	not legacy support label
OBJ-PAIR-GAP	$g_{\text{pair}}^{\text{str}}$	ideal pair-space gap	not mass or confidence gap
OBJ-MASS-GAP	$g_{\text{mass}}^{\text{str}}$	true structural support gap	not pair gap or pole value
OBJ-OBS-MASS-GAP	$\widehat{g}_{\text{mass}}^{\text{str}}$	reconstructed interval gap	not true support gap
OBJ-SEP	$s_{\text{car,ref}}$	common-record separator	SPIC-conditional
OBJ-REC-RAD	r_{rec}	record confidence radius	not structural radius

C.2 Handoff pullback

The section functors $s_i : \mathcal{A}_b \rightarrow \mathcal{H}_i^b$ and common identity functor $j_b : \mathcal{A}_b \rightarrow \mathcal{I}_b$ carry natural isomorphisms $\epsilon_i : \pi_i s_i \Rightarrow j_b$. The capstone handoff is

$$\mathcal{H}_8^b = \mathcal{H}_6^b \times_{\mathcal{I}_b}^h \mathcal{H}_7^b \times_{\mathcal{I}_b}^h \mathcal{H}_{7.5}^b.$$

The comparison cells in a pullback object are isomorphisms between the three images in $\mathcal{I}_b$; no natural transformation is asserted between π_i having different domains.

C.3 Descended-map equations

$$\begin{aligned}
q_{\text{phys}} F_{\text{phys}} &= \overline{F}_{\text{phys}} q_{\text{src}}, & q_{\text{SCIC}} F_{\text{scic}} &= \overline{F}_{\text{scic}} q_{\text{phys}}, \\
q_{\text{term}} F_{\text{term}} &= \overline{F}_{\text{term}} q_{\text{acc}}, & q_{\text{obs}} F_{\text{obs}} &= \overline{F}_{\text{obs}} q_{\text{acc}}, \\
q_{\text{pert}} \text{Pr}_{\text{pert}} &= \overline{\text{Pr}}_{\text{pert}} q_{\text{frozen}}.
\end{aligned}$$

The terminal and observable maps are parallel consequences of the accepted apex. The record equation $q_{\text{rec}}^p F_{\text{rec}}^p = \overline{F}_{\text{rec}}^p q_{\text{obs}}^p$ exists only after SPIC supplies common-record congruence.

D Empirical inputs, information overlap, and parameter status

D.1 Classification of empirical uses

Use ID	Record	Role	Authority note
EU-01	oscillation source/detection evidence	Anchor	forms the base target
EU-02	KATRIN KNM1-5	Anchor / calibration	input, not PCR8 validation
EU-03	combined ^{76}Ge search response	Anchor / calibration	null search is not held out
EU-04	D3 HNL/LFU/CLFV records	Anchor / derivation	overlaps later HFLAV use
EU-05	D4 public mass-sum record	Anchor	complete likelihood remains NotReady
EU-06	D5 symmetry/charge equations	Structural derivation	not an empirical selector
EU-07	electron-active process tester	Derivation / calibration	already upstream
EU-08	ATLAS prompt-HNL response grid	Benchmark / screen	not full-source recast
EU-09	HEPData Figure 4a ordinate replay	Compatibility screen	not likelihood non-exclusion
EU-10	HFLAV Tau-2023 ratios	Compatibility check (overlap)	overlaps D3 LFU information
EU-11	separate mW normalization screen	Screening only	not epsilon_weak95
EU-12	future flavor-complete active/heavy CCST packet	Future held-out	requires protocol identity, SPIC, frozen nuisance
EU-13	future independent LNV protocol	Future held-out	nonzero can exclude exact-global; null does not select
EU-14	complete cosmological likelihood	Future NotReady	not part of current child theorem
EU-15	future complex-pole normalization and lifetime response	Future NotReady	requires all X7/X8 certificates

D.2 Information-overlap consequences

The principal nonindependence edges are:

- D3 LFU $\rightarrow$ HFLAV compatibility: no independent validation claim;
- D3 active tester $\rightarrow$ PCR-7 light normalization: derivation/calibration use;
- ATLAS response grid $\leftrightarrow$ ordinate replay: one benchmark information family;
- ^{76}Ge D2 $\rightarrow$ any same-record LNV reuse: held-out eligibility requires a new independent protocol;
- oscillation anchors $\rightarrow$ future CCST: broad domain overlap is allowed only because the future use is a predeclared joint relation absent from target formation, with disjoint record ancestry.

D.3 Parameter status and permitted use

Object	Value/status	Authority	Blocked promotion
$\epsilon_{\text{row}} = \delta_E = s_E^2$	$0 < \epsilon_{\text{row}} \leq 1/256$	Target-qualified bound	not an independently measured weak-rate deficit
$H(I_E)$	symbolic source-qualified domain	Structural exact	not a numerical interval

Object	Value/status	Authority	Blocked promotion
[28, 30] GeV	D3 source-certified scouting band	Search benchmark	not predicted VJ pole mass
$c\tau_A^{\text{resp}} = 0.2$ mm	exact ATLAS reweight-grid coordinate	Search benchmark	not physical proper lifetime or pp laboratory mean
$p = 0.1349$	frozen HFLAV compatibility statistic	Compatibility check	not independent validation or epsilon information
$\epsilon_{95}^{\text{scr}} \simeq 1.62 \times 10^{-3}$	leading-response mW screen	Screening only	not epsilon_weak95
$\epsilon_{95}^{\text{weak}}$	NotReady	NotReady	not replaced by screen
$g_{\text{pair}}^{\text{str}}$	$\sqrt{6}(s_E - \bar{\eta}_V)^2$ lower bound	Structural exact	not mass gap or detector confidence
$g_{\text{mass}}^{\text{str}}$	symbolic true light-heavy support gap	Conditional envelope	not pair gap or pole value
$\widehat{g}_{\text{mass}}^{\text{str}}$	observed reconstructed interval gap	Conditional envelope	not true support gap
$s_{\text{car,ref}}$	positive SPIC record separator	Conditional envelope	not inferred from the ideal pair-space gap without certified transport
r_{rec}	attainable record confidence radius above floor	Conditional envelope	not structural or mass radius
$\text{Pole}_H^{\text{phys}}$	$\det D^{-1} = 0$ envelope	Conditional envelope	no numerical value without certificates

E Semantic trace and counterexample conditions

The semantic capstone equivalence is fixed before the trace:

$$Y_1 \equiv_{\text{cap}} Y_2 \iff \nu(Y_1) = \nu(Y_2) \text{ with equal definedness for all } \nu \in \text{Use}_{\text{cap}}^b.$$

The derived trace is

$$\text{Tr}_8(Y) = (A_Y, P_Y, T_Y, O_Y, U_Y, R_Y; C_Y),$$

where the object factors are target/source, physical apex, terminal interface, observable packet, perturbation/update, and protocol/record. Its relation package is the exact finite set

$$C_Y = \{C_{AP}, C_{PT}, C_{PO}, C_{TOI}, C_{PU}, C_{TU}, C_{OU}, C_{UR}, C_{AR}, C_{\partial}, C_{\Delta}\}.$$

Here C_{PU} and C_{TU} are the physical- and terminal-under-perturbation action cells. The final cell states their compatibility with C_{PT} , C_{PO} , C_{TOI} , C_{OU} , the record leg, and the boundary, including the induced higher cocycle; it is not a catch-all relation. A rival preserving the marginals but failing one cell is a `CrossInterfaceCoherenceFailure`, not a new standing authorizer.

For a rival Y , let $\text{Diff}_8(Y)$ be the set of trace fields and cells differing from the frozen capstone trace. P8-11A derives

$$\text{Diff}_8(Y) = \emptyset \iff Y \equiv_{\text{cap}} Y_{\text{frozen}}.$$

A faithful same-scope counterexample preserves one ready-child target/source, one-apex physicality, terminal and ROC, observable trace, perturbation/update typing, protocol/record semantics, the child/parent boundary, and every coherence diagram, then contradicts a stated conclusion. It reopens the minimal false premise set; it is not excluded by definition.

F Robustness domains and SPIC conditions

Feature	Domain	Status	Guard
One carrier unperturbed	all accepted ready-child X	exact	PCR-6 terminal theorem
Positive carrier radius	declared P75-13 cell	cell-certified	fixed support/faces/ROC
Closure	U_{iso}	class-exact	block isometry
Rank three	U_{rank}	positive radius	$\eta_V < s_E$
Joint core	$U_{\text{joint}}^b(X)$	intersection theorem	carrier, isometry, rank
Degeneracy	central symmetry stratum	protected	zero unrestricted radius
Universality	equal-singular-value stratum	protected	zero unrestricted radius
Pair gap	$D_{\text{pair}} \subseteq U_{\text{iso}} \cap U_{\text{rank}}$	exact	$\bar{\eta}_V < s_E$
Record selection	common-record SPIC	conditional	separator, inverse, coverage

F.1 Distinct guard objects

$$g_{\text{pair}}^{\text{str}} \geq \sqrt{6}(s_E - \bar{\eta}_V)^2$$

is an ideal pair-space theorem. Mass resolution uses

$$r_m < g_{\text{mass}}^{\text{str}}/4, \quad g_{\text{mass}}^{\text{str}} > 2r_m, \quad \widehat{g}_{\text{mass}}^{\text{str}} > 2r_m.$$

Record selection uses

$$2r_{\text{rec}} < s_{\text{car,ref}}$$

and the two inverse-radius, nuisance, floor, coarsening, identifiability, and coverage guards. No equality among these objects is licensed.

G Benchmark, weak normalization, and complex-pole requirements

G.1 Search-benchmark authority chain

1. PCR-5D3 supplies a source-certified $M \in [28, 30]$ GeV continuum with positive response margins in its frozen specialization.
2. PCR-7.5 retains $H(I_E)$ as the symbolic VJ source domain and refuses to identify it with the D3 band.
3. 30 GeV is a direct ATLAS mass anchor; 28 GeV is supported by the source continuum, not direct simulation.
4. 0.2 mm is an exact discrete response-reweight coordinate, not a full-source proper lifetime.
5. The HEPData ordinate replay is a simplified-model compatibility screen, not a likelihood or source-complete recast.
6. Tagged on-shell- W kinematics and prompt/near-prompt reconstruction make the pair a useful initial scouting point.

G.2 Weak-record stop rule

The frozen rule is

$$\epsilon_{95}^{\text{weak}} < \frac{1}{256} \implies \text{publish a tightened source-weak intersection;}$$

otherwise retain the marked-child bound and report compatibility only. The available public reconstruction did not supply a jointly covered physical $\epsilon_{95}^{\text{weak}}$. The second alternative therefore applies. The screen $\epsilon_{95}^{\text{scr}} \approx 1.62 \times 10^{-3}$ is retained as a diagnostic number and may not be substituted into the first branch.

G.3 X7 complex-pole continuation fields

Field	Required content
Inverse propagator	complete matrix-valued $D^{-1}(z)$ on a declared analytic domain
Continuation	sheet, branch, threshold, and orientation prescription
Self energy/width	complete finite-order or controlled nonperturbative kernel
Noncommutativity	treatment of noncommuting mass and width matrices
Pole identity	unordered pole set with multiplicity/collision rules
Coherent response	response object when individual poles are not identifiable
Normalization	nonempty source/weak normalization intersection
Status	conditional until all fields are supplied

G.4 X8 physical mass/lifetime localization fields

Field	Required content
Independent record	not D3 prompt-lifetime calibration and not upstream child formation
Response cover	polewise-identifiable and coherent-response branches
Compatible types	exhaustive finite source/response type atlas
Nuisance envelope	joint detector, theory, acceptance, and background covariance
Mass bridge	source coordinate to physical pole/record relation
Input scheme	electroweak normalization and response scheme
Inversion	exact set-valued or certified higher-order rule near singular conditioning
Laboratory map	proper lifetime plus boost, angle, acceptance, and reconstruction convolution
Status	numerical mass and lifetime NotReady

H Falsifier structure, held-out outcomes, and open questions

H.1 Minimal failure set

For a qualified record R on the induced claim poset,

$$\text{MinFail}(R) = \{v \in \text{Fail}(R) : \nexists u \in \text{Fail}(R), u \prec_8 v\}.$$

The set may contain more than one node. A protocol failure is reported before any physical failure set; a parent record lies outside the ready-child poset.

H.2 Held-out outcome schema

Status	Record condition	Scientific report
CarrierPacketMatched	carrier inverse contains truth; reference inverse empty	target-qualified support, no unique origin
ReferencePacketMatched	reference inverse contains truth; carrier inverse empty	falsify or update ready-child packet
PartialFeatureFailure	qualified coordinate outside class	report full $\text{MinFail}(R)$
BothInverseRegionsEmpty	neither frozen family covers record	new response/source or protocol defect
ProtocolNotQualified	SPIC, independence, freeze, or coverage fails	no physical verdict
InconclusiveAtResolution	separator/guards not achieved	preserve frozen prediction

H.3 Open questions

ID	Open burden	Status	Exact consequence
OB8-01	universal cover of every accepted realization by P75-13 certified trace-transport cells	NotReady	positive perturbative carrier radius remains cell-qualified
OB8-02	instantiated experiment-specific SPIC	NotReady	robust detector selection remains conditional
OB8-03	qualified held-out CCST data above frozen threshold	AwaitingData	no empirical validation status
OB8-04	complete contemporaneous immutable history of all formation choices	EvidenceBounded	historical no-influence claim not fully certifiable
OB8-05	joint weak covariance endpoint $\epsilon_{95}^{\text{weak}}$	NotReady	retain 1/256 source bound; weak record is compatibility check
OB8-06	X7/X8 complete complex-pole kernel and compatible types	NotReady	physical pole mass and lifetime remain NotReady
OB8-07	complete source proper lifetime and pp boost/acceptance convolution	NotReady	0.2 mm remains response-chart benchmark
OB8-08	complete PCR5E parent neutrino closure	NotReady	positive theorem remains child-qualified
OB8-09	D3-1.1 Global Fibre Completion	NotReady	no transport from D3-1.3 children
OB8-10	complete D4 joint cosmological likelihood JLR-D4-1	NotReady	public-summary child cannot be promoted to complete cosmology
OB8-11	present-facility full protocol, acceptance and exposure certification	NotReady	finite-protocol existence is not current facility readiness
OB8-12	cross-fermion FCGS source/role comparison	NotReady	no quark/charged-lepton universality
OB8-13	unique microscopic/UV origin across continuum child moduli	OpenByTheorem	one bearer within a realization does not select one realization

I Consistency conditions and evidentiary scope

ID	Condition	Requirement
RG8-01	direct authority	final versions and ARC mandate fixed
RG8-02	child/parent scope	all positive claims child-qualified
RG8-03	handoff pullback	source-bound object and arrow actions, cells, and universal property
RG8-04	dependency reconciliation	scientific formation and proof-use structures agree; graph acyclic; ambient mathematics separate
RG8-05	object identity	pair, mass, observed, separator, and radius objects distinct
RG8-06	quotient descent	congruence and commuting equation for every map
RG8-07	one-apex coherence	all eleven explicit cross-interface relations commute
RG8-08	endpoint erasure	source and endpoint-erasure signatures agree before the triangles are formed

ID	Condition	Requirement
RG8-09	empirical firewall	use-level roles and overlaps exact
RG8-10	parameter authority	numerical promotions blocked
RG8-11	rival trace	independent semantic-use class, descent, soundness, saturation, and eleven-relation trace completeness
RG8-12	robustness/gaps	feature domains and five gap/radius objects separated
RG8-13	falsifiers	atomic feature-claim graph and qualified minimal-failure antichain
RG8-14	open burdens	no promotion of NotReady conclusions
RG8-15	claim fidelity	every claim traces to its theorem and scope
RG8-16	evidentiary separation	reproducibility evidence agrees with, but does not substitute for, the proofs

The supplementary records describe the handoff pullback, scientific dependencies, ambient mathematical authorities, source-signature and endpoint-erasure comparison, quotient equations, semantic capstone-use class, six-layer and eleven-relation trace, atomic feature-claim graph, gap distinctions, empirical uses, parameter authority, rival classes, robustness domains, benchmark and pole boundaries, held-out conditions, and open burdens. Their mutual agreement supports reproducibility and scope control. It does not prove the imported physics, future detector SPIC, or empirical truth.

J Supplementary scope correspondence and reproducibility

The following statements preserve the internal PCR-8 identifiers used by the dependency record. They concern the correspondence between the mathematical results and their presentation; they are supplementary scope statements, not additional physical theorems.

Supplementary statement P8-24. Correspondence between claims and scope.

Every physics-facing statement in the title, abstract, principal theorem, interpretation, and conclusion is read with the domains and boundaries specified in P8-22 and P8-23. Joint robustness is asserted only on $U_{\text{joint}}^b(X)$; pair, mass, observed-interval, record-separator, and confidence-radius objects remain distinct; and failure localization uses a minimal antichain.

Proof.

Comparison with the endpoint matrix excludes parent universalization, promotion of a search benchmark to a pole prediction, identification of structural support with a physical complex pole, conflation of pair-space separation with a mass gap or confidence radius, replacement of complete trace identity by marginal rival checks, and any claim that a reporting priority is a unique scientific cause. These are scope restrictions on the stated results.

AbstractFidelityClosed_g.

Supplementary statement P8-25. Reproducibility and evidentiary separation.

The manuscript and its supplementary records give separately inspectable descriptions of the scientific dependency graph, proof uses, ambient mathematical authorities, upstream source signatures, endpoint-erasure comparison, handoff functors, semantic capstone-use class, capstone diagram, atomic claim graph, quotient relations, and authority distinctions. Agreement among these descriptions establishes internal consistency and reproducibility; it neither replaces the manuscript proofs nor proves the imported physics.

Proof.

The dependency descriptions are compared only after separate preparation. The same discipline is applied to source-normal-form signatures and endpoint-erasure inputs, to the three handoff functors and their naturality witnesses, and to the semantic-use class and its derived trace. The atomic feature graph is checked for acyclicity and for correspondence with every falsifier label. Quotient identities, gap distinctions, SPIC conditions, open questions, and claim-language boundaries are compared with the statements in which they arise. Negative controls test whether any one description has been obtained by copying the conclusion of another. These comparisons establish the integrity of the scholarly record, not the empirical truth of the imported physical results.

ArtifactAuditClosed₈.

Supplementary statement P8-26. Declared scope and downstream use.

Within the declared two-child carrier and prediction problem, P8-23 completes the composition result. The shorthand

PCR8Final_{child-qualified}.

denotes only this local closure. Downstream work may evaluate held-out data, instantiate SPIC, complete X7/X8, construct a present-facility analysis, reopen the parent, or formulate a cross-fermion target. Any change to the typed handoff object, dependency structure, source-signature analysis, semantic capstone-use class, atomic claim graph, packet, benchmark authority, reference family, open-question status, or falsifier lattice defines a new target or a revised prediction.

Proof.

P8-23 gives the complete positive result, and P8-22 states its open boundary. P8-24 records the correspondence between the public claims and those results, while P8-25 records reproducibility without evidentiary substitution. The local composition requirements are therefore discharged without supplying a parent theorem, an instantiated SPIC, a numerical X7/X8 pole, or an empirical outcome.

J.1 Data and reproducibility

The supplementary project contains the source and records needed to inspect the authority assignments, handoff and quotient relations, dependency structure, endpoint-erasure comparison, empirical-use classifications, robustness domains, falsifier graph, and open boundaries. It also reproduces the typeset article. These materials document the analysis; scientific authority remains with the stated arguments and their cited sources.

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